

Embedded Systems and IoT

Ingegneria e Scienze Informatiche - UNIBO

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[module-1.3]

SENSORS AND ACTUATORS - OVERVIEW

OUTLINE

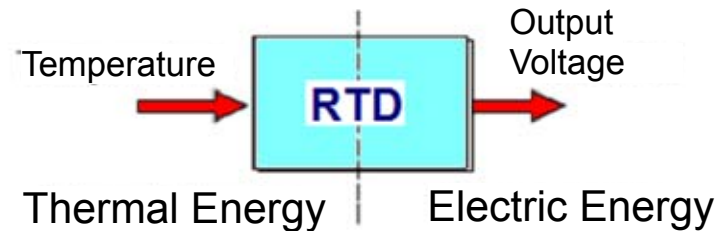
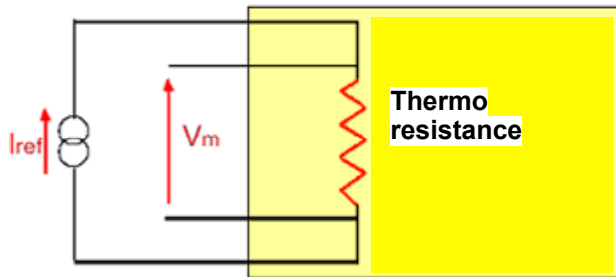
- This module provides an overview of main concepts about sensors and actuators, focussing in particular on those that will be used in assignments
- Outline
 - Sensors and actuators: intro
 - Sensors and input devices - typologies
 - Actuators and output devices - typologies
 - Driving devices of external circuits

SENSORS AND ACTUATORS

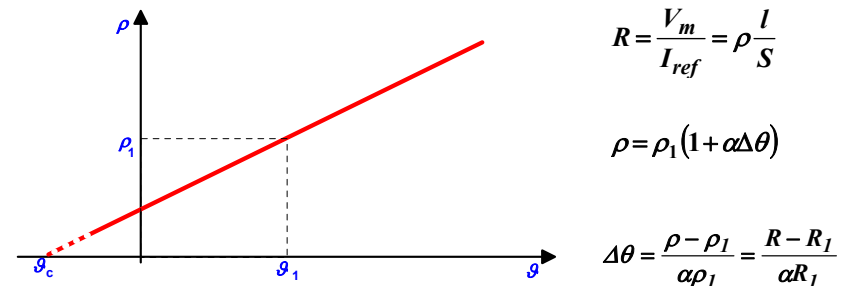
- An embedded systems interact with external environment using sensors and actuators
- **Sensors**
 - ***transducer device*** that allow to *measure* some physical phenomenon (e.g., temperature, humidity, radiations, etc), or to detect or quantify chemical concentrations (e.g. gas)
 - it provides a measurable representation of a phenomenon in some measuring scale or interval
 - e.g.: voltage
 - can be **analog** (if the electrical value produced in output - e.g. voltage or current - is continuous) or **digital** (discrete set of values)
 - to convert analog signals, microcontrollers have a specific device included called analog-digital-converter (ADC)
- **Actuators**
 - devices that produce some effect on the environment
 - analog or digital (equipped of a DAC, Digital Analog Converter)

SENSORS: WORKING PRINCIPLES

- Based on physical laws that rule the relationships between *physical quantities* to be measured and an *output electrical quantity* (e.g. voltage)
- Example: thermistor (temperature sensor)



- Resistivity changes according to temperature
- To get the signal, an input current is needed, to measure then an output voltage



<http://ww2.unime.it/ingegneria/new/materiale/sensori2.pdf>

PHYSICAL QUANTITIES AND SIGNALS

- Physical quantities measured by sensors can be classified as:
 - **continuous**
 - continuous values ranging in a certain range
 - e.g.: temperature, distance, angular speed of a motor, etc
 - **discrete**
 - can assume only a discrete set of values
 - es. rotation direction, number of pieces processed, etc
- The information associated to physical quantities are called *signals*
 - **analog signals**
 - **discrete signals**
 - *logical signals* — only two admissible values
 - *codified signals* — if more than two values

SENSORS TAXONOMY

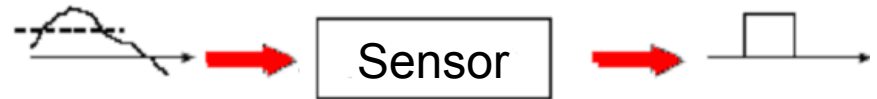
- **Analog sensors**

- output continuous electrical signal



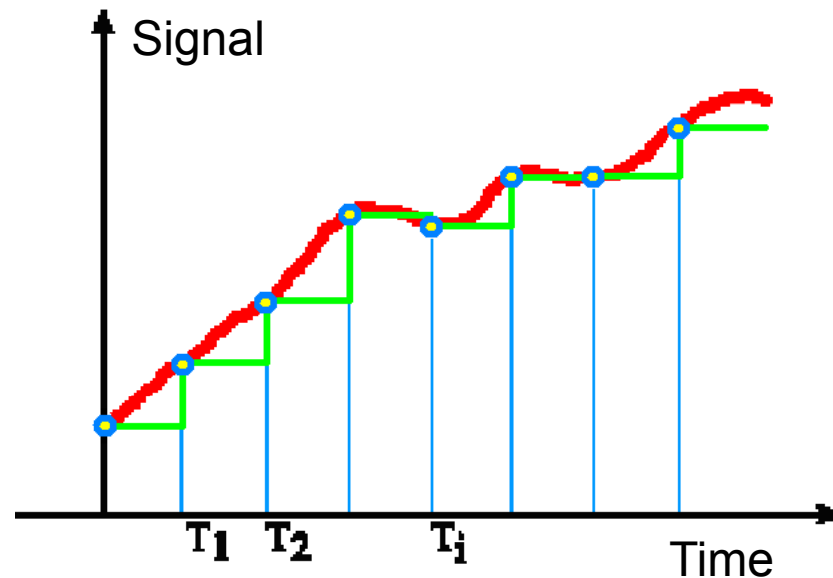
- **Digital sensors**

- output: digital information at some resolution
- **logical sensor**
 - boolean output
- **codified sensor**
 - output: string of bits



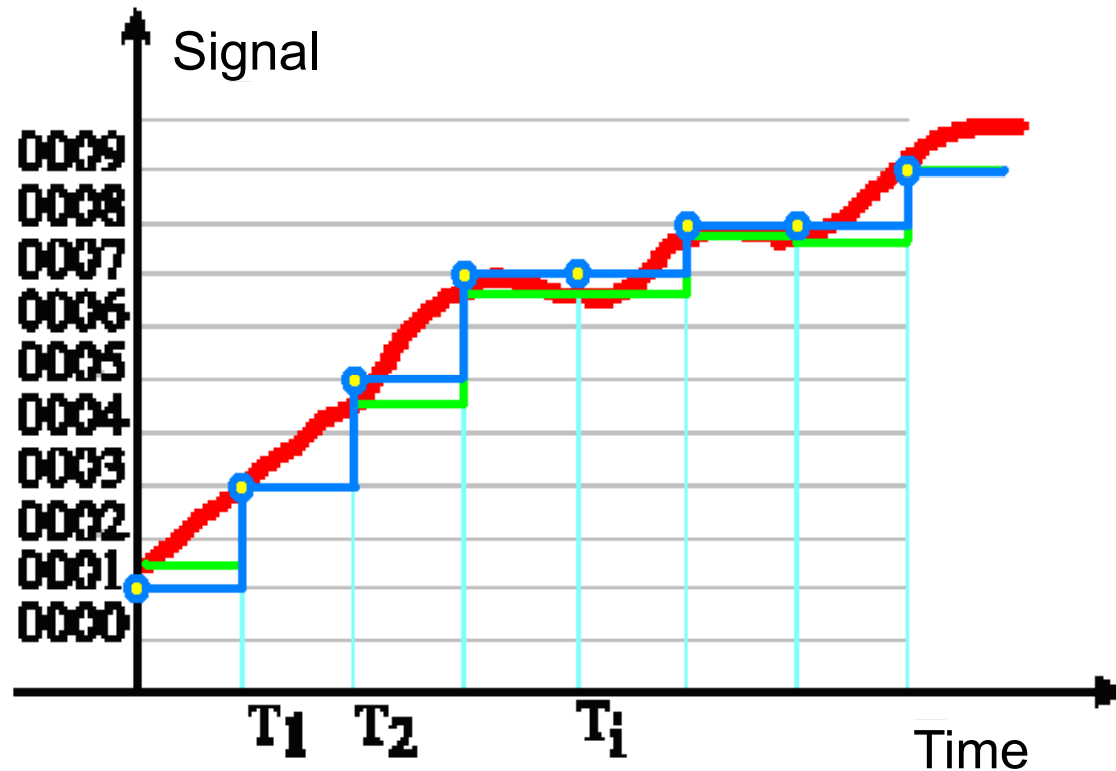
SIGNAL SAMPLING

- An analog signal produced by an analog sensor to be processed by a digital computer must be sampled by means of an analog to digital converter device (ADC)
 - output: digital signal
- **Signal sampling**
 - reading signal samples at specified discrete times



SIGNAL QUANTISATION

- Sampled signal is approximated to the nearest digital value



MEASURING AND MEASURES

- **Physical quantity *measurement*** = comparison between two quantities of the same physical property (homogeneous)
 - it establishes the relationship between some unknown quantity and another known one, taken as a *reference*
 - This operation is called *measurement*
- **SI** (“Système international d’unités”) system: most widespread unit of measurement system
 - seven fundamental physical quantities and related units of measurement

SI - UNIT OF MEASUREMENT SYS.

Unit name	Unit symbol	Quantity name	Definition (incomplete) ^[n 1]	Dimension symbol
metre	m	length	Original (1793): $\frac{1}{10\,000\,000}$ of the meridian through Paris between the North Pole and the Equator.^{FG} Interim (1960): 1 650 763.73 wavelengths in a vacuum of the radiation corresponding to the transition between the 2p¹⁰ and 5d⁵ quantum levels of the krypton-86 atom. Current (1983): The distance travelled by light in a vacuum in $\frac{1}{299\,792\,458}$ second.	L
kilogram ^[n 2]	kg	mass	Original (1793): The grave was defined as being the weight [mass] of one cubic decimetre of pure water at its freezing point.^{FG} Current (1889): The mass of the International Prototype Kilogram (Le Grand K).^[38]	M
second	s	time	Original (Medieval): $\frac{1}{86\,400}$ of a day. Interim (1956): $\frac{1}{31\,556\,925.9747}$ of the tropical year for 1900 January 0 at 12 hours ephemeris time. Current (1967): The duration of 9 192 631 770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the caesium-133 atom.	T
ampere	A	electric current	Original (1881): A tenth of the electromagnetic CGS unit of current. The [CGS] electromagnetic unit of current is that current, flowing in an arc 1 cm long of a circle 1 cm in radius, that creates a field of one oersted at the centre.^[39] IEC Current (1946): The constant current which, if maintained in two straight parallel conductors of infinite length, of negligible circular cross-section, and placed 1 m apart in vacuum, would produce between these conductors a force equal to 2×10^{-7} newtons per metre of length.	I
kelvin	K	thermodynamic temperature	Original (1743): The centigrade scale is obtained by assigning 0 °C to the freezing point of water and 100 °C to the boiling point of water. Interim (1954): The triple point of water (0.01 °C) defined to be exactly 273.16 K.^[n 3] Current (1967): $\frac{1}{273.16}$ of the thermodynamic temperature of the triple point of water	Θ
mole	mol	amount of substance	Original (1900): The molecular weight of a substance in mass grams.^{ICAW} Current (1967): The amount of substance of a system which contains as many elementary entities as there are atoms in 0.012 kilogram of carbon-12.^[n 4]	N
candela	cd	luminous intensity	Original (1946): The value of the new candle is such that the brightness of the full radiator at the temperature of solidification of platinum is 60 new candles per square centimetre. Current (1979): The luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 5.4×10^{14} hertz and that has a radiant intensity in that direction of $\frac{1}{683}$ watt per steradian.	J

DERIVED UNITS

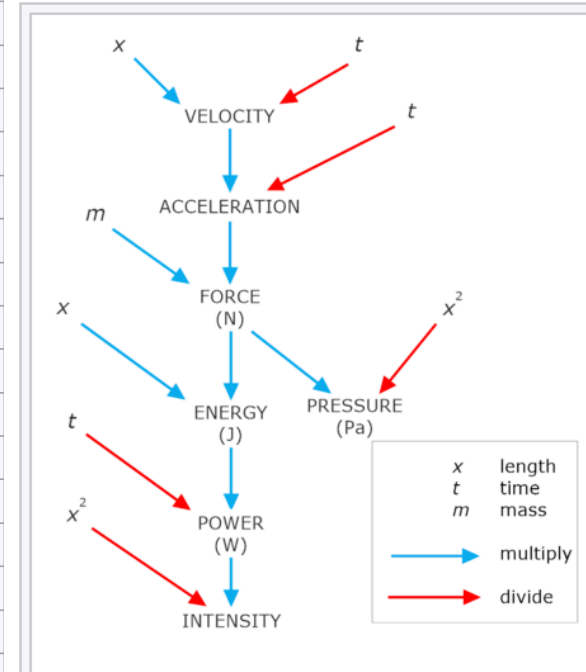
SI derived units with special names and symbols^{[2]:15}

Name	Symbol	Quantity	In SI base units	In other SI units
radian ^[N 1]	rad	plane angle	m/m	1
steradian ^[N 1]	sr	solid angle	m ² /m ²	1
hertz	Hz	frequency	s ⁻¹	
newton	N	force, weight	kg · m · s ⁻²	
pascal	Pa	pressure, stress	kg · m ⁻¹ · s ⁻²	N/m ²
joule	J	energy, work, heat	kg · m ² · s ⁻²	N · m = Pa · m ³
watt	W	power, radiant flux	kg · m ² · s ⁻³	J/s
coulomb	C	electric charge	s · A	
volt	V	electric potential, voltage, emf	kg · m ² · s ⁻³ · A ⁻¹	W/A = J/C
farad	F	capacitance	kg ⁻¹ · m ⁻² · s ⁴ · A ²	C/V = C ² /J
ohm	Ω	resistance, impedance, reactance	kg · m ² · s ⁻³ · A ⁻²	V/A = J · s/C ²
siemens	S	electrical conductance	kg ⁻¹ · m ⁻² · s ³ · A ²	Ω ⁻¹
weber	Wb	magnetic flux	kg · m ² · s ⁻² · A ⁻¹	V · s
tesla	T	magnetic flux density	kg · s ⁻² · A ⁻¹	Wb/m ²
henry	H	inductance	kg · m ² · s ⁻² · A ⁻²	Wb/A
degree Celsius	°C or °C	temperature relative to 273.15 K	K	
lumen	lm	luminous flux	cd · sr	cd · sr
lux	lx	illuminance	cd · sr · m ⁻²	lm/m ²
becquerel	Bq	activity referred to a radionuclide (decays per unit time)	s ⁻¹	
gray	Gy	absorbed dose (of ionising radiation)	m ² · s ⁻²	J/kg
sievert	Sv	equivalent dose (of ionising radiation)	m ² · s ⁻²	J/kg
katal	kat	catalytic activity	mol · s ⁻¹	

Notes

1. ^{a b} The radian and steradian are defined as dimensionless derived units.

Screenshot



Arrangement of the principal measurements in physics based on the mathematical manipulation of length, time, and mass.

PREFIXES

Prefix		Base 10	Decimal	English word		Adoption ^[nb 1]	Etymology	
Name	Symbol			Short scale	Long scale		Language	Source word
yotta	Y	10 ²⁴	1 000 000 000 000 000 000 000 000	septillion	quadrillion	1991	Latin	eight ^[nb 2]
zetta	Z	10 ²¹	1 000 000 000 000 000 000 000	sextillion	trilliard	1991	Latin	seven ^[nb 2]
exa	E	10 ¹⁸	1 000 000 000 000 000 000	quintillion	trillion	1975	Greek	six
peta	P	10 ¹⁵	1 000 000 000 000 000	quadrillion	billiard	1975	Greek	five ^[nb 2]
tera	T	10 ¹²	1 000 000 000 000	trillion	billion	1960	Greek	four, monster ^[51]
giga	G	10 ⁹	1 000 000 000	billion	milliard	1960	Greek	giant
mega	M	10 ⁶	1 000 000	million		1873	Greek	great
kilo	k	10 ³	1 000	thousand		1795	Greek	thousand
hecto	h	10 ²	100	hundred		1795	Greek	hundred
deca	da	10 ¹	10	ten		1795	Greek	ten
		10 ⁰	1	one		–		
deci	d	10 ^{−1}	0.1	tenth		1795	Latin	ten
centi	c	10 ^{−2}	0.01	hundredth		1795	Latin	hundred
milli	m	10 ^{−3}	0.001	thousandth		1795	Latin	thousand
micro	μ ^[nb 3]	10 ^{−6}	0.000 001	millionth		1873	Greek	small
nano	n	10 ^{−9}	0.000 000 001	billionth	milliardth	1960	Greek	dwarf
pico	p	10 ^{−12}	0.000 000 000 001	trillionth	billionth	1960	Spanish	peak, a little bit
femto	f	10 ^{−15}	0.000 000 000 000 001	quadrillionth	billiardth	1964	Danish	fifteen, Fermi ^[nb 4]
atto	a	10 ^{−18}	0.000 000 000 000 000 001	quintillionth	trillionth	1964	Danish	eighteen
zepto	z	10 ^{−21}	0.000 000 000 000 000 000 001	sextillionth	trilliardth	1991	Latin	seven ^[nb 2]
yocto	y	10 ^{−24}	0.000 000 000 000 000 000 000 001	septillionth	quadrillionth	1991	Latin	eight ^[nb 2]

1. [^] Prefixes adopted before 1960 already existed before SI. The introduction of the [CGS system](#) was in 1873.

2. [^] [a b c d e](#) Part of the beginning of the prefix was modified from the word it was derived from, e.g.: "peta" (prefix) vs "penta" (source word).

3. [^] The abbreviation mcg is often used in medicine instead of μg to prevent confusion with mg, which can lead to overdoses. A lower case u is often used instead of μ where the latter character is unavailable.

4. [^] The *fermi* was introduced earlier with the symbol "fm", which prompted the reinterpretation of the "f" as a prefix to "m", with *femto* being derived from the Danish word *femten*.

MEASUREMENT UNCERTAINTY

- Multiple measurements of the same parameter typically don't produce the same values
 - various causes, related to the measurement tools, environments
- To describe this dispersion of values, the concept of **measurement uncertainty** is introduced
 - can be interpreted as the level of confidence with respect to the result,
 - a physical quantity can be determined only at some specified finite level of uncertainty (i.e. the **error**)
 - uncertainty quantifies the imprecision or inaccuracy of the measurement
- ***The measurement of some physical quantity:
value $\pm$ uncertainty (error) and unit of measurement***

MEASUREMENT ERRORS

- **Systematic errors**
 - an error is systematic if, once the experimental conditions are fixed, it systematically impacts in the same way (same quantity and sign) on the measurements
- **Random (accidental) errors**
 - errors whose influence (quantity and sign) can change as the measurements are repeated
 - environment conditions can be the source of these errors
- **“Gross” errors**
 - errors done by either the operator or some failure of the measuring instruments

METROLOGICAL CHARACTERISTICS OF MEASURING INSTRUMENTS

- **Static characteristics**

- about operating conditions in which the input varies slowly, measuring then the output
 - *Accuracy, Precision, Linearity, Sensitivity, Resolution, Repeatability, Reproducibility, Stability, ...*

- **Dynamic characteristics**

- about operating conditions in which the input varies fast, to check possible output attenuations and delay
 - the dynamic characteristics of a sensor could be critical since it may constraint the whole bandwidth of the whole control system
 - response time

STATIC CHARACTERISTIC

- A **transducer** static characteristic can be represented by a function:

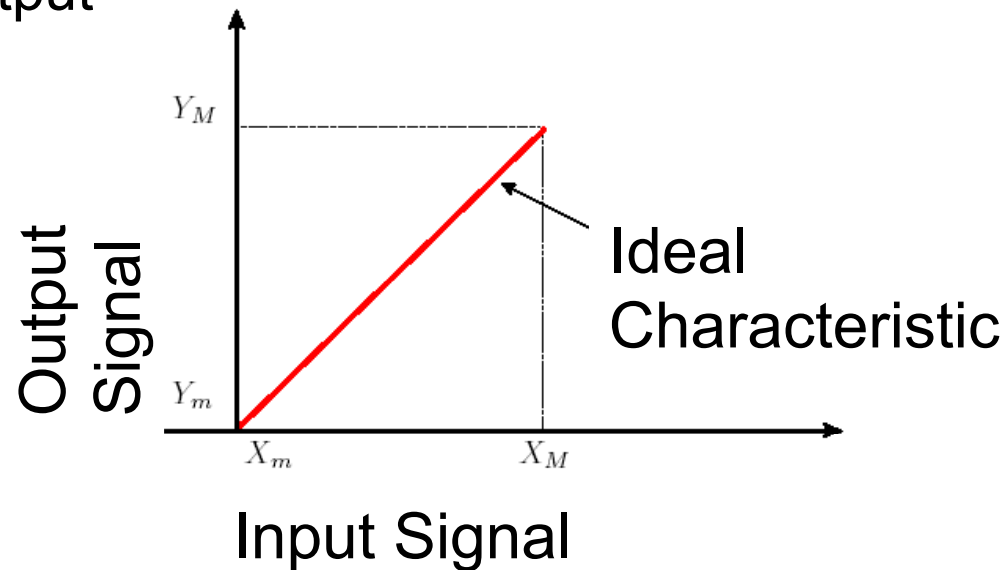
$$Y = f(X)$$

where

- X is the input signal
- Y is the output signal
- This static characteristic is defined in a finite range called input range between X_m and X_M , and has output on the output range or span between Y_m and Y_M .
 - Input range: $X_s = X_M - X_m$
 - Output range: $Y_s = Y_M - Y_m$

SENSOR GAIN

- The static characteristic of sensor should have ideally a linear pattern
 - **sensor gain** (K) = proportionality constant between input and output



- commercial transducer has a real static characteristic which can be different from the ideal one
 - depends on the quality of the sensor

LINEARITY

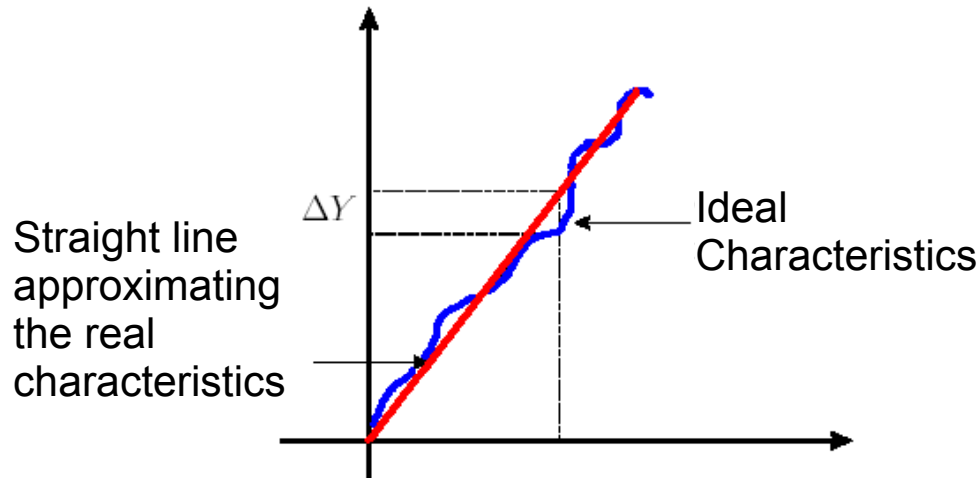
- Linear transducer => the function describes a straight line:

$$Y = K X$$

where K is the gain

LINEARITY ERROR

- Linearity error = max deviation of the transducer output compared to the linear characteristics better approximating the real one



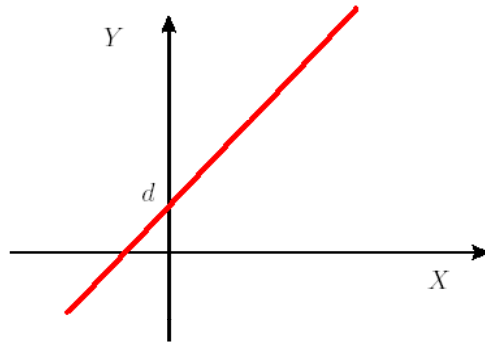
- Linear characteristics can be obtained using the least squares method
 - finding the straight line $f(X)$ minimising: $J = \sum_{i=0}^N [r(X_i) - Y_i]^2$
- If ΔY is the maximum delta, then the non linearity percentage error can be computed as:

$$e_L\% = \frac{\Delta Y}{Y_M - Y_m} 100$$

OFFSET AND THRESHOLD ERRORS

- *Offset* error is the d output value when the input is zero

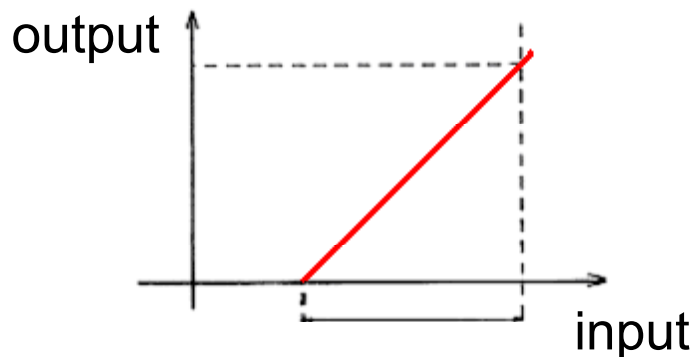
$$Y = f(X) = KX + d$$



To have a linear characteristic, the “offset” d must be removed:

$$Y = f(X) - d$$

- Threshold error is about the lowest possible input that can be measured — that could be not zero



To have a linear characteristic, the threshold s must be removed:

$$Y = f(X - s)$$

OTHER ERRORS

- **Gain error**

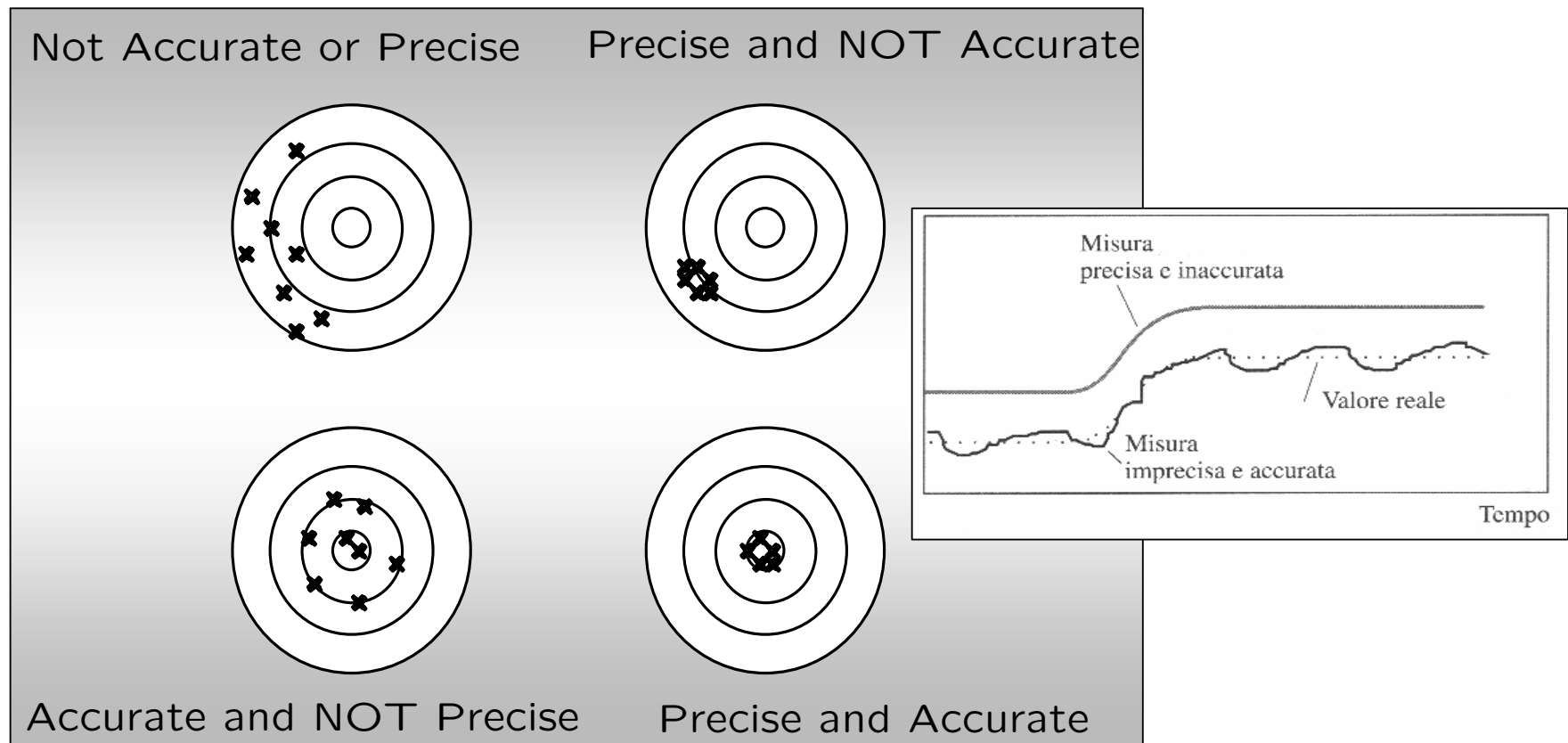
- Difference between the ideal characteristic (K) and the straight line (K_1) better approximating the real characteristic of the transducer
- Gain error (a percentage):

$$e_G\% = \frac{|K_1 - K|}{K} 100$$

- **Quantisation error**

- Sampling operation
=> quantisation
=> error (approximation) on the sampled value

ACCURACY AND PRECISION



ACCURACY

- Sensor (measurement) accuracy — a measurement of how much the value read by a sensor is different from the real/correct value
 - it can be computed as the maximum difference between *the measurement* (or “read”) provided by the sensor and the correct value of the measured physical quantity

Accuracy as full scale percentage	$\epsilon_f = 100 \cdot \frac{X_m - X_v}{X_{FS}}$
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- X_v = measurement with no errors
- X_m = actual measurement
- X_{FS} = max value considering the range

Accuracy as percentage of the measurement	$\epsilon_a = 100 \cdot \frac{X_m - X_v}{X_v}$
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Absolute Accuracy	$\epsilon(X_v) = X_m - X_v $
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EXAMPLE

- A pressure sensor measuring values in the 0-10 bar range, with an accuracy error of $\pm 1.0\%$ *compared to the full scale* (10 bar)
- Then, the max error is $10 \times 1.0\% = 10 \times 0.01 = 0.1$ bar

PRECISION

- The **precision** measures how much a sensor is affected by *random errors*
 - if we do a large number of measurements with a high-precision sensor, then the range of these values will be small
- Precision is not related to accuracy, but *repeatability*
 - reproducing the same or similar output values for the same input, in the same operating conditions
- It can be computed as deviation of measured values with respect to the average value

EXAMPLE

- The size of a room is measured by a sonar 10 times, reporting the following measurements: 5.381, 5.379, 5.378, 5.382, 5.380, 5.383, 5.379, 5.377, 5.380, 5.381
- The measurement done by another reference instrument reports 5.374m (assumed as true value)
- Precision
 - average measurement value: 5.380
 - max deviation (under the value): $5.377 - 5.380 = -0.003$
 - max deviation (above the value): $5.383 - 5.380 = +0.003$
 - > la precisione del sensore è esprimibile come ± 0.003 m
- Max accuracy error = $5.383 - 5.374 = +0.009$ m

CALIBRATION

- Low accuracy => systematic errors or *bias*
- **Calibration** (*calibration*) of a sensor
 - tuning of sensor parameters to align the output values to those correct as measured with the reference instrument

DYNAMIC BEHAVIOUR

- Dynamic behaviour of a sensor
 - the response to an input is not instantaneous
 - it is often negligible wrt the overall process dynamics
- When we apply a step input signal, the signal response varies until stabilising to the new value
- Timings
 - **rise time**
 - time taken to go from 10% to 90% of the final value
 - **response time**
 - time taken to achieve a percentage of the final value

SENSORS - SOME TYPOLOGIES

SENSORS: TYPOLOGIES

- Proximity sensors
- Movement detection sensors
- Angular position transducers
- Acceleration sensor
- Contact sensors
- Optical sensors
- Electricity and magnetism sensors
- Environmental parameter sensors
- Physiological/vital parameter sensors
- Sound sensors
- Localisation sensors
- Identity detection sensors

LARGE MARKET

- Large availability on the marker
- E.g. Adafruit sensor list
 - <https://www.adafruit.com/category/35>

WIRING LIBRARIES AND ARDUINO PLAYGROUND/WIKI

- **Wiring libraries about sensors and actuators**
 - <http://wiring.org.co/learning/libraries/>
- **Arduino project hub**
 - examples, projects
 - <https://create.arduino.cc/projecthub>

PROXIMITY

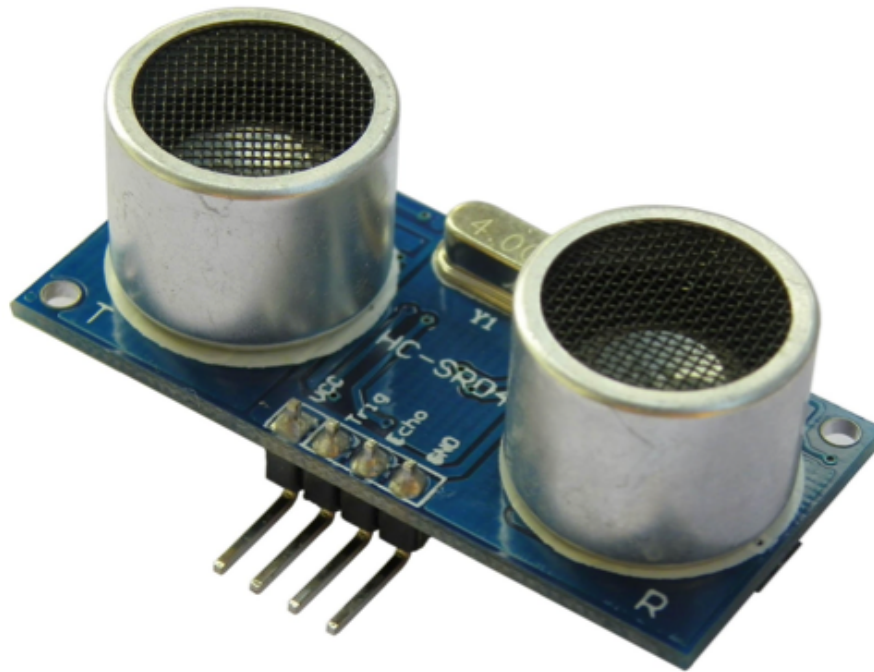
- **Proximity sensors** can detect the presence of some object at some distance
 - max distance = rated or nominal distance or range
- Based on different technologies
 - ultrasonic sensors (sonars)
 - optical sensors
 - inductive sensors
 - magnet sensors
 - ...
- Typically proximity sensors just detect an object
 - without measuring the distance

ULTRASONIC PROXIMITY SENSORS

- Sonar operating principle
 - they emit an sonar impulse (240 KHz), and detect the echo back — in the case of an object in the nominal range
- Pros
 - quite long nominal distances (up to 10 m);
 - no affected by electromagnetic interference
 - can detect object of different materials (but sound-adsorbing material)
 - can detect objects without any specific setup or marks on the object side
- Cons
 - slow response time
 - can be affected by the object surfaces (orientation)

HC-SR04 EXAMPLE

- **HC-SR04** sensor
 - datasheet and docs:
https://docs.google.com/document/d/1Y-yZnNhMYy7rwhAgyL_pfa39RsB-x2qR4vP8saG73rE/edit

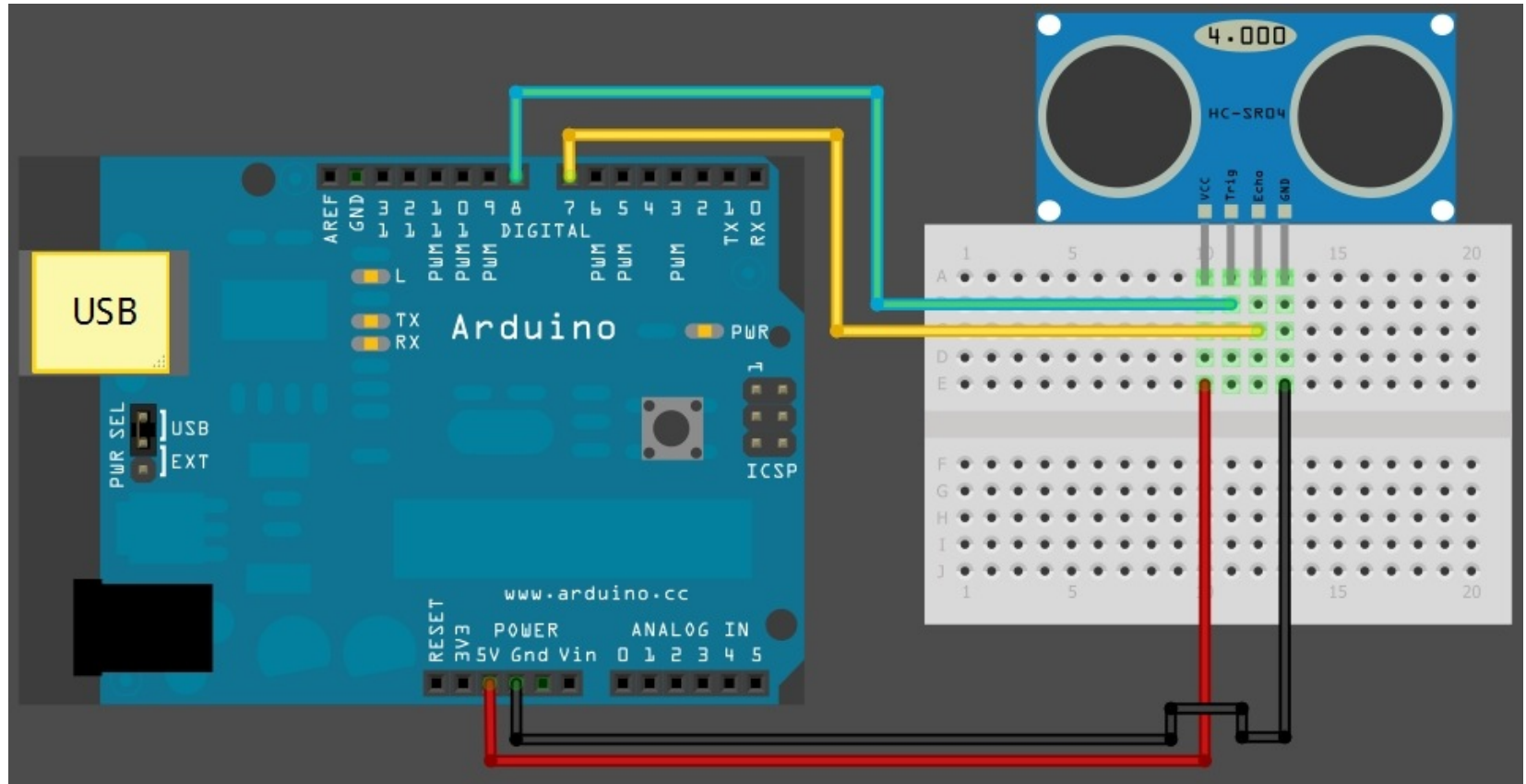


USAGE EXAMPLE IN ARDUINO

- HC-SR04 - Characteristics
 - detection distance: from 3 cm to 6 meters
 - 2 control pins: **Echo** and **Trig**
- Usage strategy / operating principles:
 - *an impulse is sent on the Trig pin and we measure time elapsed to receive the echo on Echo pin (if an object is present)*
 - *given that time => we can compute the distance, knowing the speed of sound (330 m/s)*
- Calculation
 - $vs * dt = 2 * d$
 - $d = (vs * dt) / 2$
 - $dt = 2 * d / vs$
- So an object at a distance of 50 cm = 0.5 m is detected in $dt = 1/330 = 0.003030 \text{ s} = 3.030 \text{ ms} = 3030 \text{ us}$
- The speed of sound depends on different environmental factors, such as the temperature: $vs = (331.3 + 0.606 * T)$
 - At 20 C degrees => 343 m/s.

USAGE EXAMPLE IN ARDUINO

- [example from Make:Sensors]
 - Trig/echo connected to pin 8/7



PROGRAM

```
const int trigPin = 8;
const int echoPin = 7;
/* supposing to have a temperature
   of 20 °C */
const float vs = 331.5 + 0.6*20;

void setup()
{
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

float getDistance()
{
  /* sending impulse */
  digitalWrite(trigPin, LOW);
  delayMicroseconds(3);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(5);
  digitalWrite(trigPin, LOW);

  /* receiving the eco */
  float tUS = pulseIn(echoPin, HIGH);
  float t = tUS / 1000.0 / 1000.0 / 2;
  float d = t*vs;
  return d;
}
...
```

```
...

void loop()
{
  float d = getDistance();
  Serial.println(d);
  delay(200);
}
```

THE ROUTINE pulseIn

- **pulseIn**
 - efficient routine part of Wiring libreria
 - monitoring the input pin where to receive the impulse, tracking the time
 - timings can be quite small (microseconds)
 - must be very efficient
- Source code
 - https://github.com/arduino/Arduino/blob/master/hardware/arduino/avr/cores/arduino/wiring_pulse.c

OPTICAL PROXIMITY SENSORS

- Based on light
 - also called photoelectric sensors
- Normally infrared light is used
 - typically not influenced by environment light
- Operating principle based on light reflection
 - strongly affected by surface orientation
 - typical distance range: from 10 to 100 cm
 - by mounting *a reflex reflector* on the object to be detected, the nominal range can arrive up to 50 m

EXAMPLE

- Sharp GP2Y0A41SK0F IR
 - infrared sensor
 - <http://www.alldatasheet.com/view.jsp?Searchword=Gp2y0a41sk0f>



MOVEMENT SENSORS

- PIR
- Angular position transducers
- tilt-ball switch
- vibration detectors

PIR - INFRARED PASSIVE SENSOR

- Electronic sensor which measures the *quantity of infrared radiation* radiated from objects in the sensor field of view — useful to detect movement
 - actually a PIR detects radiation/temperature abrupt *variations*, compared to a reference situation that the PIR takes as reference state
 - importance of calibration
- So when some object with a certain temperature enters in an (colder) zone that was previously taken as reference by the PIR, the sensor detects the variation and then the presence of the entity
 - the entrance of an object with the same temperature of the background does not trigger any detection

EXAMPLE

- PIR sensor
 - <https://learn.adafruit.com/pir-passive-infrared-proximity-motion-sensor/>



ANGULAR POSITION TRANSDUCERS

- Two types
 - one detects the position without having any specific mechanical constraint, just exploiting the gravity force
 - e.g. inclinometer
 - one detects the position given a mechanical constraint => **encoder**
 - the encoder is an electromechanical device that converts the angular position of its rotating axis into electrical pulses, further processed and converted into a digital numerical signal
- Various technologies
 - capacitive/inductive, magnetic, optical

ACCELERATION SENSORS

- **Accelerometer**
 - sensor measuring the acceleration of a moving object (equipped with the sensor)
 - different types, based on different physical operating principles
- **Gyroscope**
 - rotating device that preserves a fixed direction for its rotating axis
 - can be used for different purposes
 - e.g. as compass, pointing to north

EXAMPLE

- MPU-6050 - accelerometer + gyroscope
 - <http://playground.arduino.cc/Main/MPU-6050>



MEMS TECHNOLOGY

- Electronic accelerometers and gyroscopes are based on **MEMS** technology
 - Micro Electro-Mechanical Systems
 - system of micro devices (size ~micro-meter) of different nature (mechanical, electrical and electronic) strongly integrated in the same chip/substrate (silicon)
- Very important for modern embedded systems

CONTACT SENSORS

- **Tactile buttons and microswitch**
- **Potentiometer**
 - it allows for changing the resistance of a component — and then the voltage level at the terminals — in a mechanical way
 - electrical device which is equivalent to a *voltage divider*, with a variable resistance
- **Linear potentiometer**
 - output voltage is a linear function of the input voltage
- **Capacitive sensor**
 - based on the principle of detecting the electrical capacity of the capacitor
 - a nearby object creates a capacity detected by internal circuits

EXAMPLES

- Potentiometer ALPS 10KOhm

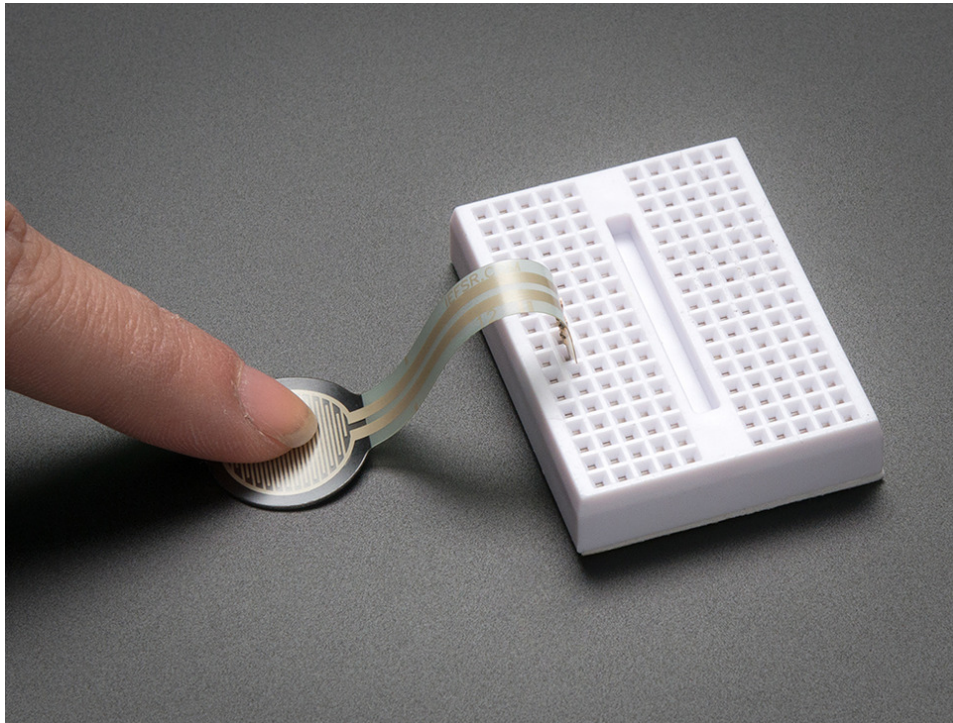


- Capacitive sensor AllLife TTP223B
 - http://www.datasheet4u.com/share_search.php?sWord=TTP223B



PRESSURE SENSOR

- Measuring the force exerted on the sensor
 - e.g.: Round Force-Sensitive Resistor (FSR)
 - <https://www.adafruit.com/products/166>



OPTICAL SENSORS

- Optical sensors or image sensors are devices converting an optical image into an electrical signal
 - e.g. in digital camera.
- Different kinds, depending on the method and technology adopted to detect colours
 - basic classification: either detecting colors or black & white (e.g. night vision)
- Operating principle
 - the image is focused on a grid of many small punctiform sensors
 - those small sensors convert light into a signal

PHOTODETECTORS

- Device detecting the electromagnetic radiation, providing a signal (current or voltage level) proportional to the intensity of the radiation
 - Different types, related to the radiation spectrum detected, minimum quantity perceived (some: single photons..)
 - also called photocells
- Varie tecnologie
 - photoresistors — based on electrons on semiconductors excited by light
 - photodiodes — based on photogenerated charge on a P-N junction
 - photovoltaic cell — similar to photodiodes but with no polarisation
 - Charge Coupled Device (CCD), Charge Injection Device (CID) e CMOS: integrate circuits based on photo generated charge in a semiconductor

EXAMPLE: PHOTORESISTOR

- **Photoresistor**

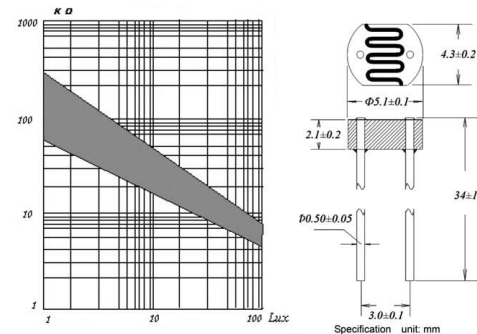
- arduino.cc/documents/datasheets/LDR-VT90N2.pdf
- resistance depends on the quantity of light perceived (inversely proportional)
 - strong light => small resistance

- **Parameters**

- R10 and R100: resistance under 10 and 100 lux
- R05: resistance being in the dark for 5 seconds

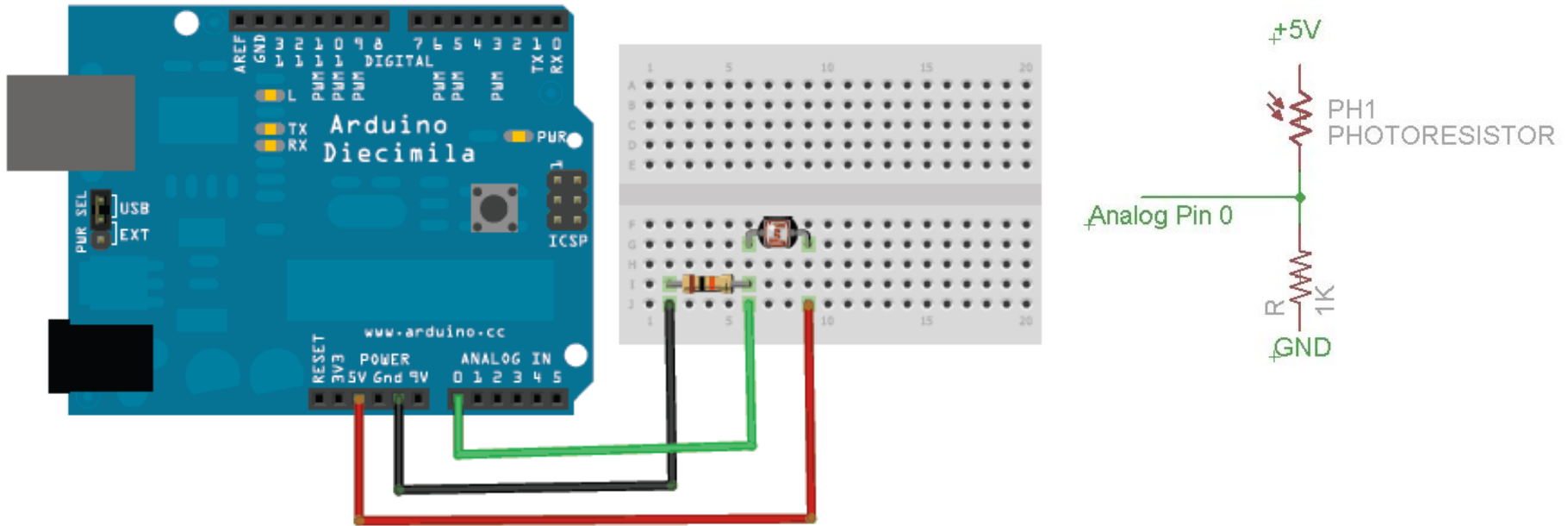
- **circuit**

- voltage divider



EXAMPLE: PHOTORESISTOR

- connecting schema: voltage divider
 - using a resistance of a known value in serie



ELECTRICITY AND MAGNETISM

- **Electric voltage and current**
 - sensors detecting the voltage level between two points of the circuit or the current flowing
 - useful whenever measuring the energy consumed by a device
 - e.g.: project <http://www.openenergymonitor.org>
- **Magnetometer**
 - measuring the magnetic field
 - either scalar (measuring only the intensity) or vectorial (measuring the intensity along a specific spatial direction)
- **Compass**
 - measuring the angle wrt magnetic north

WEATHER AND CLIMA

- **Temperature sensor**

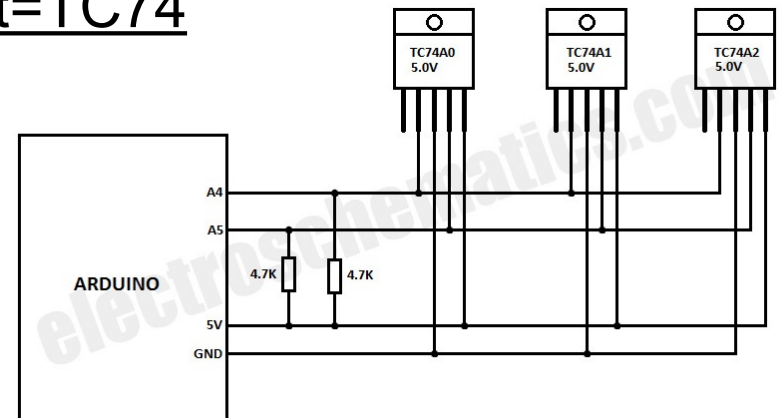
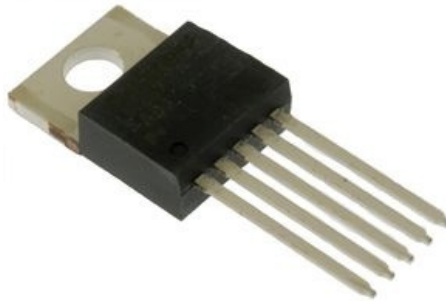
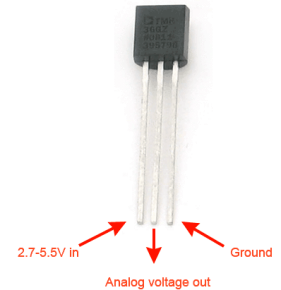
- e.g.

- analog sensor analogico TMP36

- <https://learn.adafruit.com/tmp36-temperature-sensor>

- digital sensor TC74

- <http://www.microchip.com/wwwproducts/Devices.aspx?product=TC74>



WEATHER AND CLIMA

- Humidity sensors
 - example: DHT11 Digital Temperature Humidity Sensor
 - <http://www.dx.com/p/arduino-digital-temperature-humidity-sensor-module-121350>



- Barometric sensor
 - example: BMP085 SparkFun Electronics
 - <http://www.robotstore.it/product/504/Sensore-di-Pressione-Barometrica-BMP085.html>



SENSORS DETECTING CHEMICALS

- GAS sensors
 - MQ Gas Sensor
 - <http://playground.arduino.cc/Main/MQGasSensors>



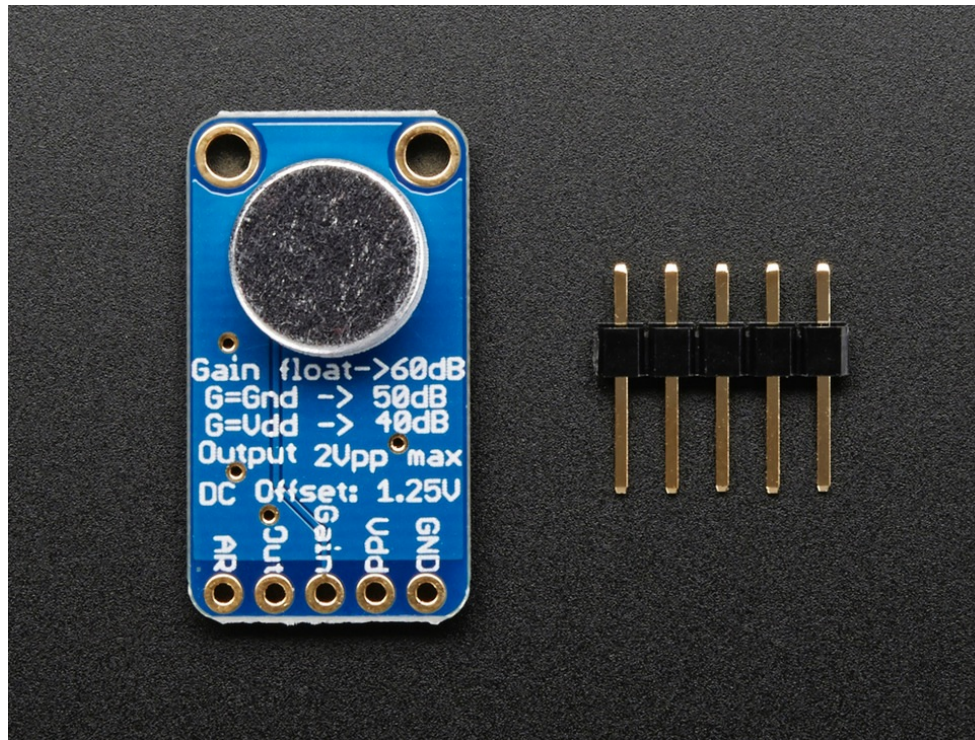
https://www.mixsensors.com/combustible/MIX2111D_CH4_Digital_Output_Natural_Gas_Detection_Module_56.html

- dust sensors | for air quality
 - e.g. Gravity: PM2.5 Air Quality Sensor



SOUND

- Microphones
 - example
 - <https://www.adafruit.com/products/1713>

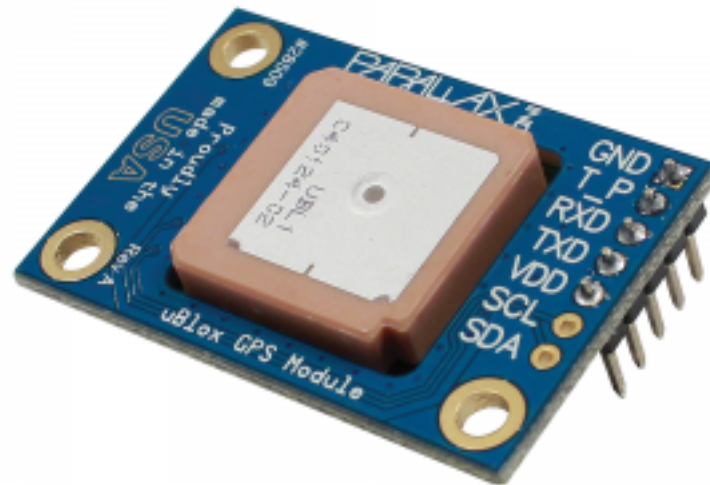


GLOBAL LOCALISATION

- **GPS Global Positioning System (GPS)**
 - positioning system and navigation based on a satellite network (32 satellites)
 - sending to GPS receivers info about coordinates and time, everywhere on earth
 - to receive info, 4 satellites must be visible
 - managed by USA
 - **accuracy level: ~meters**
 - depends on meteo condition, availability of satellites, quality of the receiver, effects on the propagation of the radio waves
- Operating principle
 - based on trilateration
 - measuring time elapsed to receive the signal from the satellites

EXAMPLE

- Parallax PAM-7Q GPS Module
 - <http://www.parallax.com/product/28509>
 - “it provides all of the standard services like location fix, speed, heading, altitude measurements, and atomic date and time”



LOCAL LOCALISATION

- **RFID**

- small tags storing data that can be read/write at distance (from centimetres to meters) by tag reader
- example: AZDelivery Kit RFID RC522

- **NFC**

- similar to RFID, but working on very small distance / contact
- example. NFC/RFID: <http://www.adafruit.com/product/789>

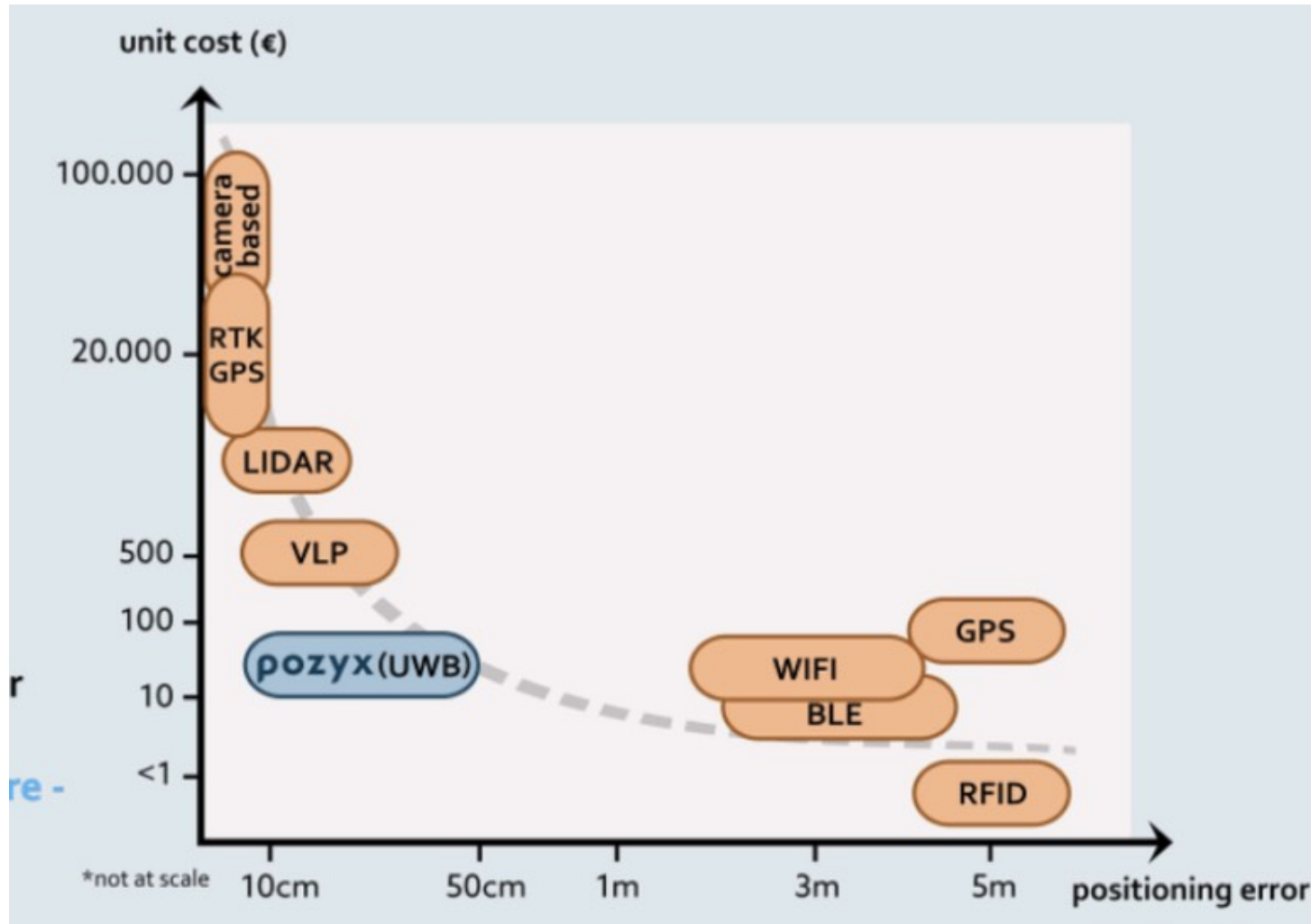
- **iBeacon**

- devices exploiting the Bluetooth technology BLE (Bluetooth low energy) to transmit a UUID in the nearby, to be received by BLE receiver
- example: <http://www.seeedstudio.com/depot/blebee-v100-p-1632.html>

- **UWB**

- based on Ultra Wide Band technology — very accurate, expensive

LOCALISATION TECHS OVERVIEW



- from : <https://www.instructables.com/10-CM-Accurate-Ultra-wide-Band-UWB-Positioning-for/>

FOR DATA INPUT

- **Keypad**
 - example: ADAFruit
 - <http://www.adafruit.com/products/419>

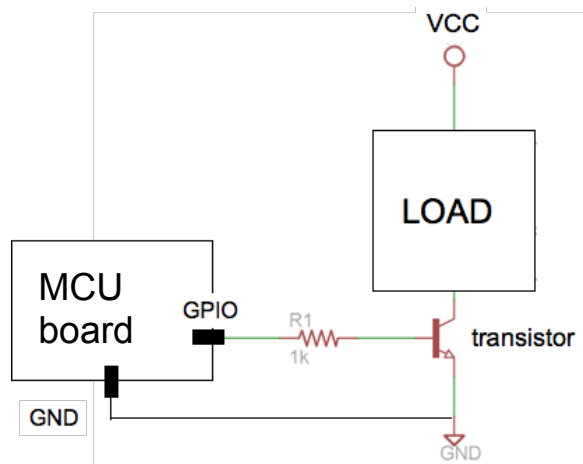
ACTUATORS AND OUTPUT DEVICES

OUTPUT TRANSDUCERS (ACTUATORS)

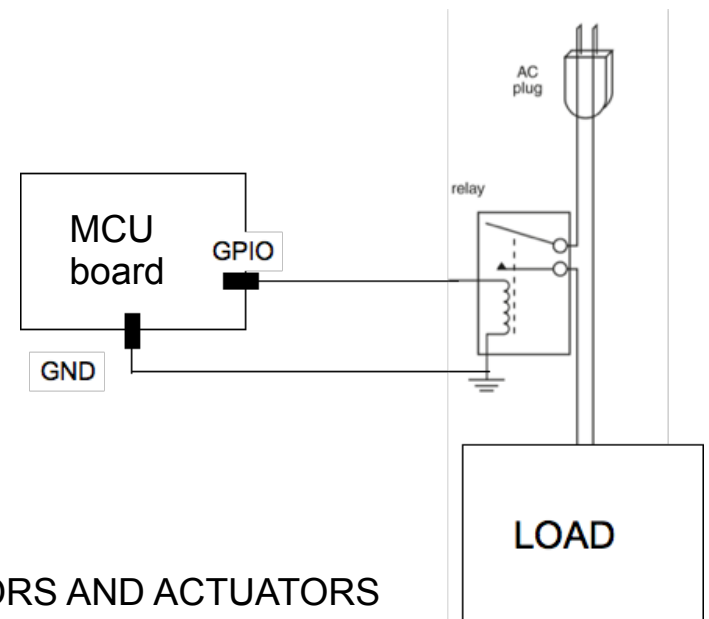
- Transducers
 - convert one kind of energy into another kind
- **Output transducers (actuators)**
 - allow embedded systems to do and control actions on the physical environment where they are embedded
 - simplest action: switching on or off a device
 - examples:
 - led, piezo buzzer, relays (switch)..

INTERFACING

- Two main cases
 - the current/voltage provided by GPIO to power the transducer is enough (i.e. 5 Volt - 20mA)
 - e.g. led, piezo
 - if it is not enough, then the device must be powered by an external circuit
 - in this case, the GPIO is used to open/close the external circuit, by means of **transistors** and **relays**, functioning as switches



ESIOT ISI-LT - UNIBO



SENSORS AND ACTUATORS

RESISTIVE VS. INDUCTIVE LOADS

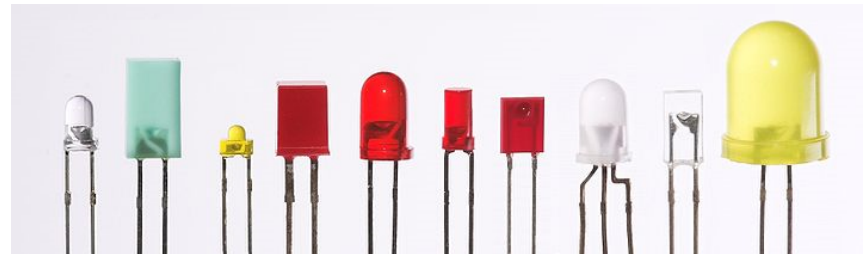
- From an electronically point of view, a controlled load can be classified in two basic categories:
 - **resistive loads**
 - can be likened to a component offering a resistance to the passage of the current, resulting in a variation of the voltage level
 - **inductive loads**
 - devices that work by inducing some current on a wire by means of the current passing on another wire or by placing the wire in a magnetic field
 - e.g.: motors, solenoids
 - *These loads generate an inverse current*
 - it is necessary to equip the circuit with protection, forcing the current to flow only in one direction

ACTUATOR EXAMPLES

- Leds
- Display LCD
- Electric motors
 - direct current, stepper, servo motor
- Speakers
 - can reproduce sounds
 - see cap. 5 [EA]
- Devices to drive power loads
 - e.g. relays, optocouplers

LEDS

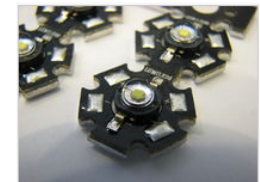
- **Light Emitting Diode**
- Opto-electronical device which exploit the optical properties of some semiconductor materials to generate photons when powered by some current
- Very widespread for their high lighting efficiency, reliability, long duration, reduced power consumption



Forme diverse di led commerciali



Led da 8, 5 e 3 mm



Un esempio di led alta potenza

DISPLAY LCD

- Display based on the optical properties of liquid crystals
 - trapped between two glass surfaces, equipped with many electrical contacts that can generate an electric field affecting the liquid
 - each electric contact control a small portion of the panel that corresponds to a pixel (or a subpixel in a color screen)
- Small power consumption
 - very good for embedded systems

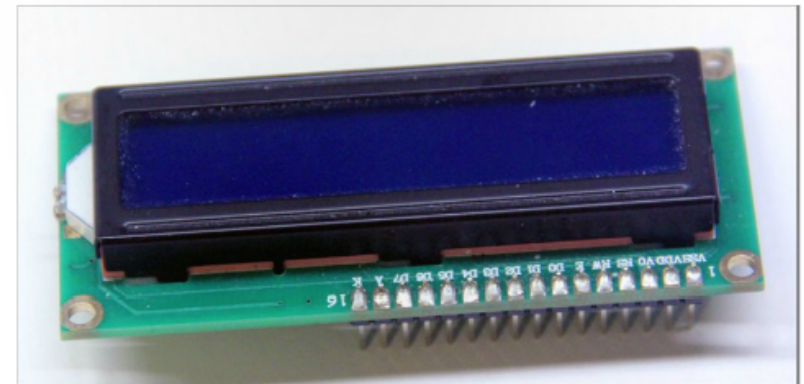
EXAMPLE

- A very common example is the parallel LCD screen 16x2 (16 columns, 2 rows), characterised by a single line of 16 pins
- All parallel LCD screen have the same pinout and can be connected to Arduino in different ways:
 - 4 pins or 8 pins
 - 8 pins mode: 6 pins must be connected to digital pins + 2 for VCC and GND
 - 6 pins mode: 4 pins are used to send the data and the other 2 for control
 - via I2C
 - es: <http://www.robot-italy.com/it/display-lcd-16-x-2-blu-i2c.html>
- Programming
 - LiquidCrystal Library provided with Arduino IDE (with I2C version)
 - many functionalities
 - cursor blinking, scrolling, customised chars creation, etc.

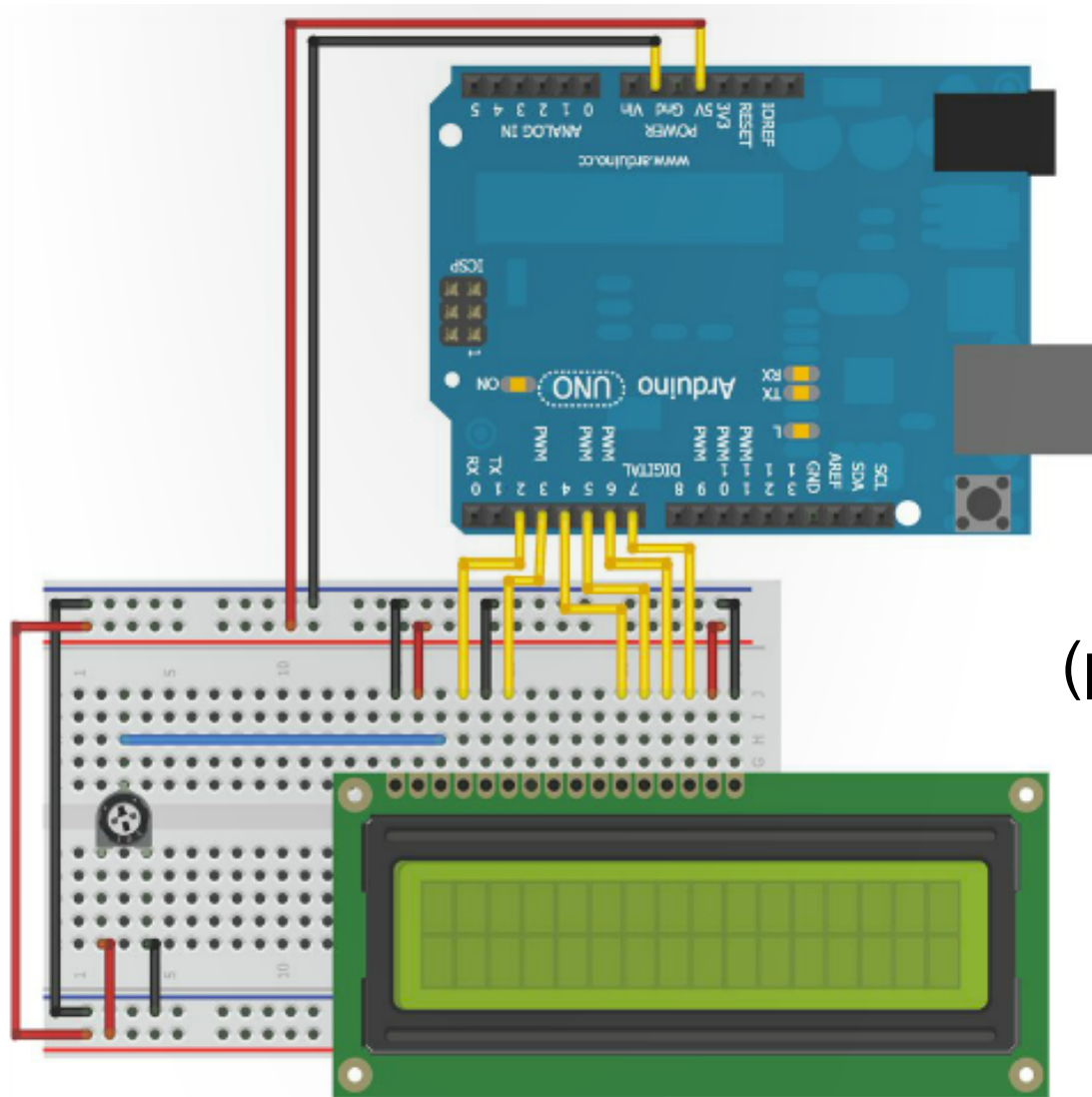
EXAMPLE

Table 10-1: Parallel LCD Pins

PIN NUMBER	PIN NAME	PIN PURPOSE
1	VSS	Ground connection
2	VDD	+5V connection
3	V0	Contrast adjustment (to potentiometer)
4	RS	Register selection (Character vs. Command)
5	RW	Read/write
6	EN	Enable
7	D0	Data line 0 (unused)
8	D1	Data line 1 (unused)
9	D2	Data line 2 (unused)
10	D3	Data line 3 (unused)
11	D4	Data line 4
12	D5	Data line 5
13	D6	Data line 6
14	D7	Data line 7
15	A	Backlight anode
16	K	Backlight cathode



EXAMPLE - FROM [EA]



(p. 203, [EA])

Image created with Fritzing.

EXAMPLE - FROM [EA] - CODE

- ```
#include <LiquidCrystal.h>

int time = 0

/* sequenza dei pin: RS EN D4 D5 D6 D7 */
LiquidCrystal lcd(2, 3, 4, 5, 6, 7);

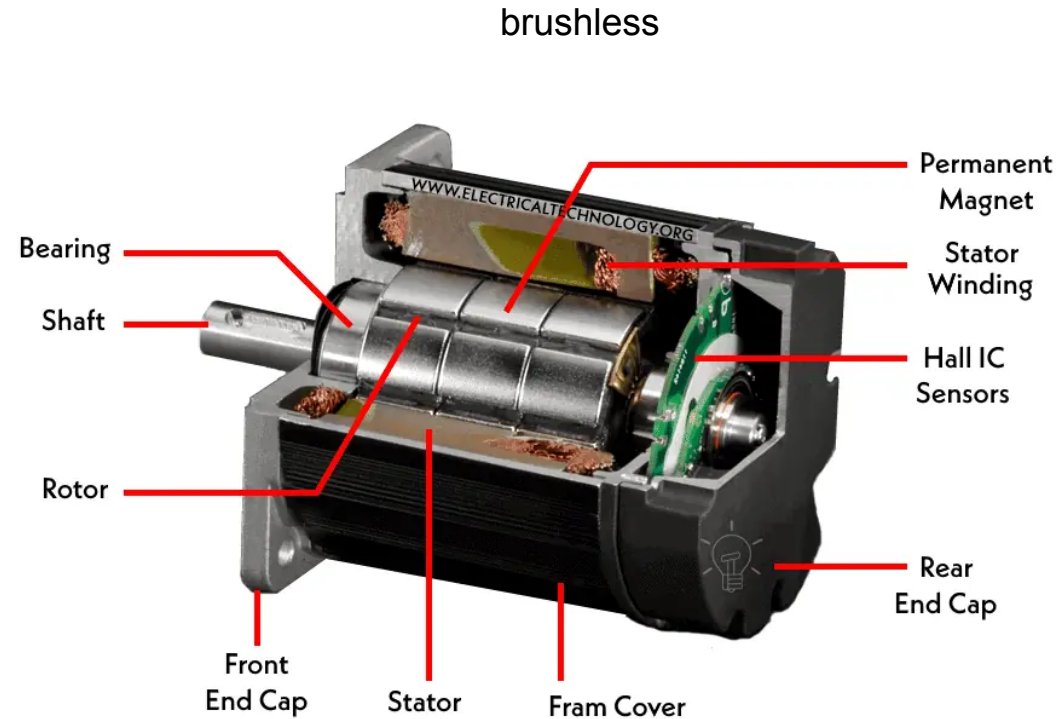
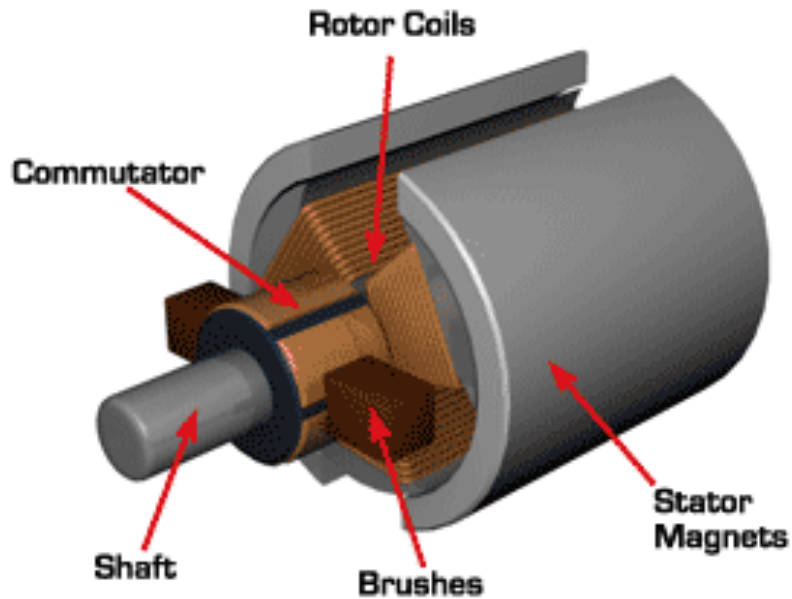
void setup()
{
 lcd.begin(16, 2); // display 16 caratteri per 2 righe
 lcd.print("Cesena");
}

void loop()
{
 lcd.setCursor(0,1);
 lcd.print(time);
 delay(1000);
 time++;
}
```

# ELECTRIC MOTORS

- **Electric motors** are electric machines that convert the input electric power into an output effect which is mechanical
- An electric motor has two main parts
  - *stator*
    - fixed/static part
  - *rotor*
    - moving part, including the shaft
- How it works
  - these two components generate a magnetic field that causes the rotor to move
- Types of electric motors used in embedded:
  - **direct current motors**
  - **stepper motors**
  - **servo motors**

# ELECTRIC MOTORS



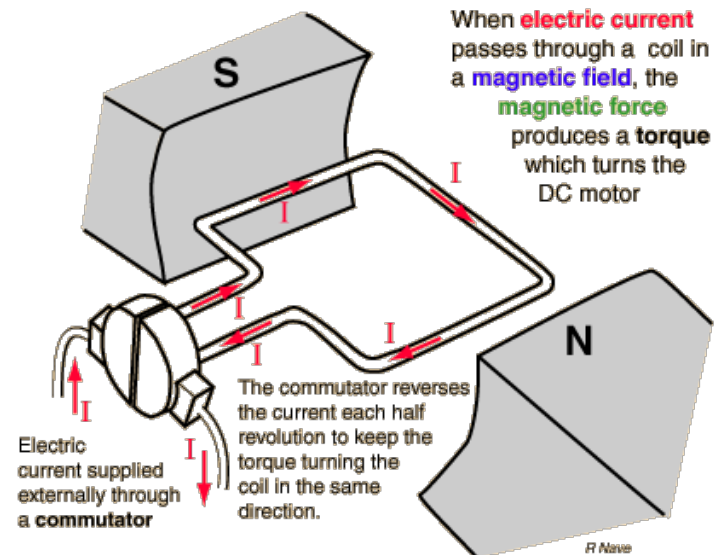
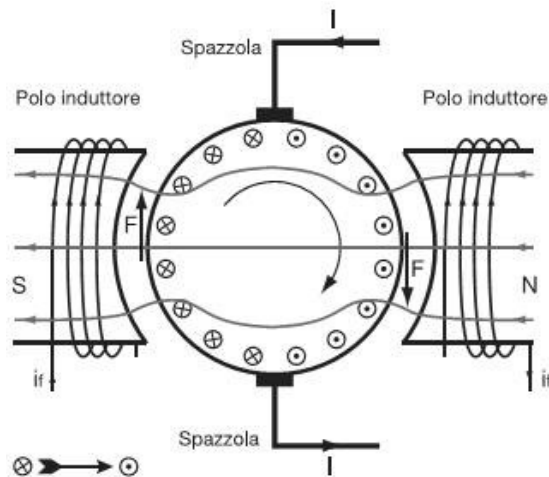
<https://www.electricaltechnology.org/2016/05/bldc-brushless-dc-motor-construction-working-principle.html>

# DIRECT CURRENT MOTORS (DC)

- Useful in projects in which motors must have
  - continuous rotation
  - full 360° movement range
  - high speed or torque
  - angular speed control
- Example
  - Robot wheels

# DC MOTORS: OPERATING PRINCIPLE

A coil of wire with a current running through it generates an **electromagnetic** field aligned with the center of the coil. The direction and magnitude of the magnetic field produced by the coil can be changed with the direction and magnitude of the current flowing through it. A simple DC motor has a stationary set of magnets in the **stator** and an **armature** with one or more windings of insulated wire wrapped around a soft iron core that concentrates the magnetic field. The windings usually have multiple turns around the core, and in large motors there can be several parallel current paths. The ends of the wire winding are connected to a **commutator**. The commutator allows each armature coil to be energized in turn and connects the rotating coils with the external power supply through brushes. (Brushless DC motors have electronics that switch the DC current to each coil on and off and have no brushes.) The total amount of current sent to the coil, the coil's size, and what it is wrapped around dictate the strength of the electromagnetic field created. The sequence of turning a particular coil on or off dictates what direction the effective electromagnetic fields are pointed. By turning on and off coils in sequence, a rotating magnetic field can be created. These rotating magnetic fields interact with the magnetic fields of the magnets (permanent or **electromagnets**) in the stationary part of the motor (stator) to create a torque on the armature which causes it to rotate. In some DC motor designs, the stator fields use electromagnets to create their magnetic fields which allows greater control over the motor.



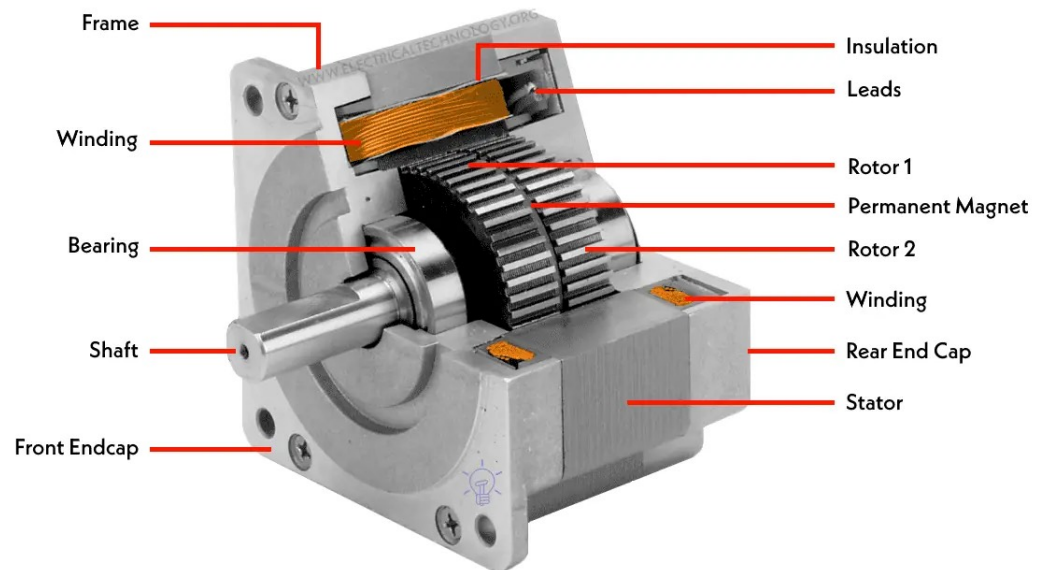
# DC MOTOR: EXAMPLE

- For Arduino projects
  - 711696 Hobby Motor - Gear
    - <http://www.robot-italy.com/it/hobby-motor-gear.html>



# STEPPER MOTOR

- Brushless dc motors in which rotation is divided in a large number of steps
  - precise control of the position of the motor shaft / rotor
- Usage
  - 3D printers
  - numerical control machines
  - arms control in rob
  - ...



<https://www.electricaltechnology.org/2016/12/stepper-motor-construction-types-and-modes-of-operation.html>

# SERVO MOTOR

- Like stepper motor, they allow for fine control of the shaft.
- Two main differences:
  - the position is absolute, not relative to the current one
  - the range is  $180^\circ$ , not  $360^\circ$
- Pros and Cons
  - easier programming
  - but limited range
- Examples (cheap)
  - 31053 HS-53 Microservo
    - <http://www.robot-italy.com/it/hs-53-microservo.html>



# SERVO MOTOR

- Servo motors can be controlled by specifying the angle of the shaft
  - between  $-90^\circ$  and  $90^\circ$
- On the device: 3 wires
  - 2 for power (nero => GND, rosso => +5V)
  - 1 for digital control
    - typically white or yellow
- **How to control a servo motor**
  - The control is done by sending a stream of digital pulses at a specified frequency
    - typically 50 pulsed per second (50 Hz)
  - the duration of the pulses determines the rotation angle
    - pulses typically go from 1 ms ( $-90$ ) to 2 ms ( $+90$ )
      - the larger is the pulse, the larger is the angle
    - linear scale — so 1.5 ms pulse => 0 degree
    - actually the real duration can greatly vary from servo to servo (to be calibrated/checked)

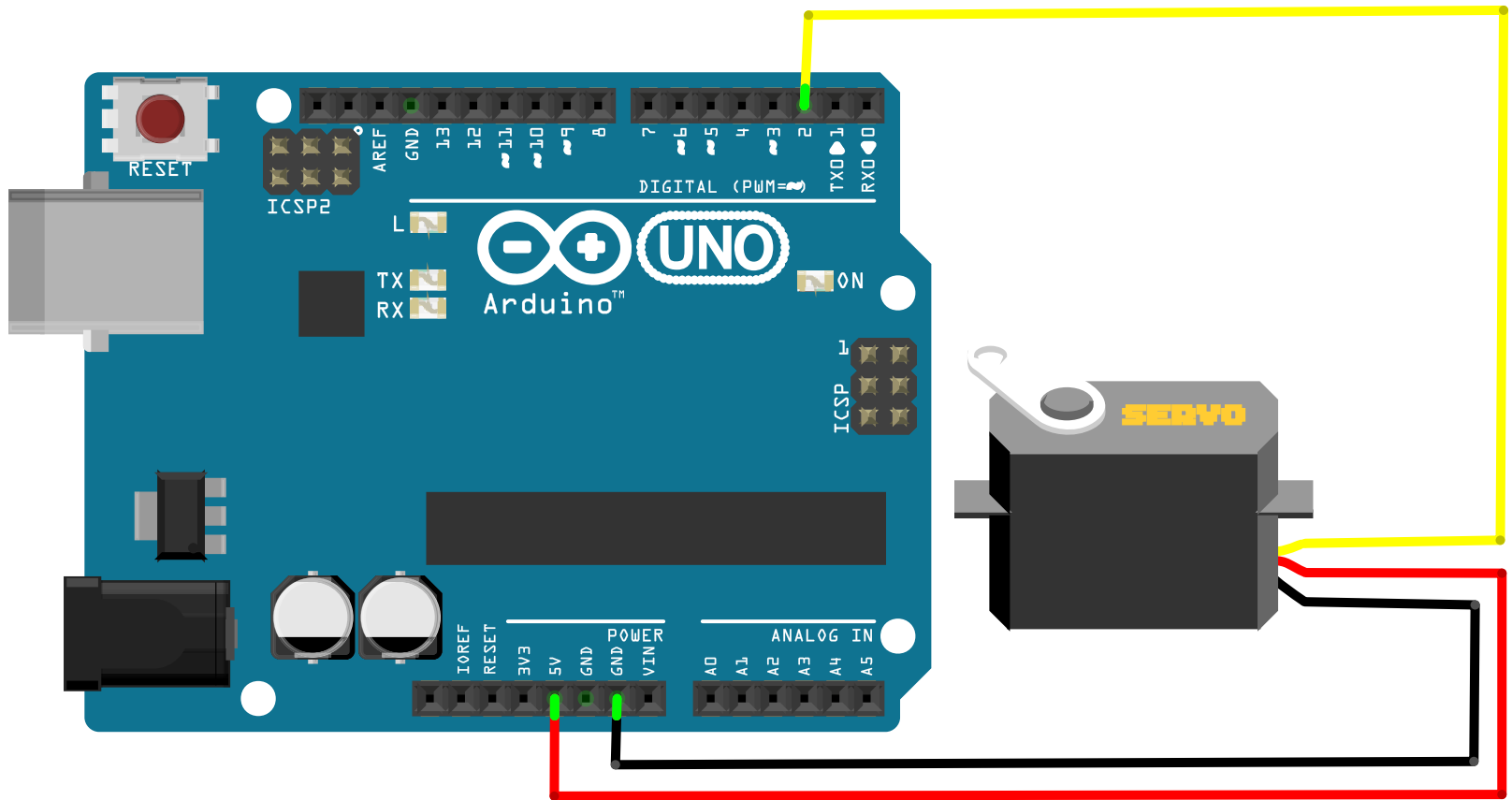
# EXAMPLE: SERVO MOTOR HS-53

- Cheap servo motor



- <http://hitecrd.com/products/servos/micro-and-mini-servos/analog-micro-and-mini-servos/hs-53-super-economy-feather-nylon-gear-servo/product>

# EXAMPLE IN ARDUINO



fritzing

# EXAMPLE IN ARDUINO

- In this case we can power directly the motor with Arduino power lines
- To control the servo we can use a digital pin
- Program example: continuous rotation from -90 to 90 and back

```
int servoPin = 2;

void pulseServo(int servoPin, int pulseLen)
{
 digitalWrite(servoPin, HIGH);
 delayMicroseconds(pulseLen);
 digitalWrite(servoPin, LOW);
 delay(15);
}

int c;
enum { MINUS_90, PLUS_90 } state;

void setup()
{
 pinMode(servoPin, OUTPUT);
 state = MINUS_90;
 Serial.begin(9600);
 c = 0;
}

void loop(){
 c++;
 switch (state) {
 case MINUS_90:
 pulseServo(servoPin, 250);
 if (c > 100){
 Serial.println("--> +90");
 state = PLUS_90;
 c = 0;
 }
 break;
 case PLUS_90:
 pulseServo(servoPin, 2250);
 if (c > 100){
 Serial.println("--> -90");
 state = MINUS_90;
 c = 0;
 }
 }
 }
}
```

# HOW IT WORKS

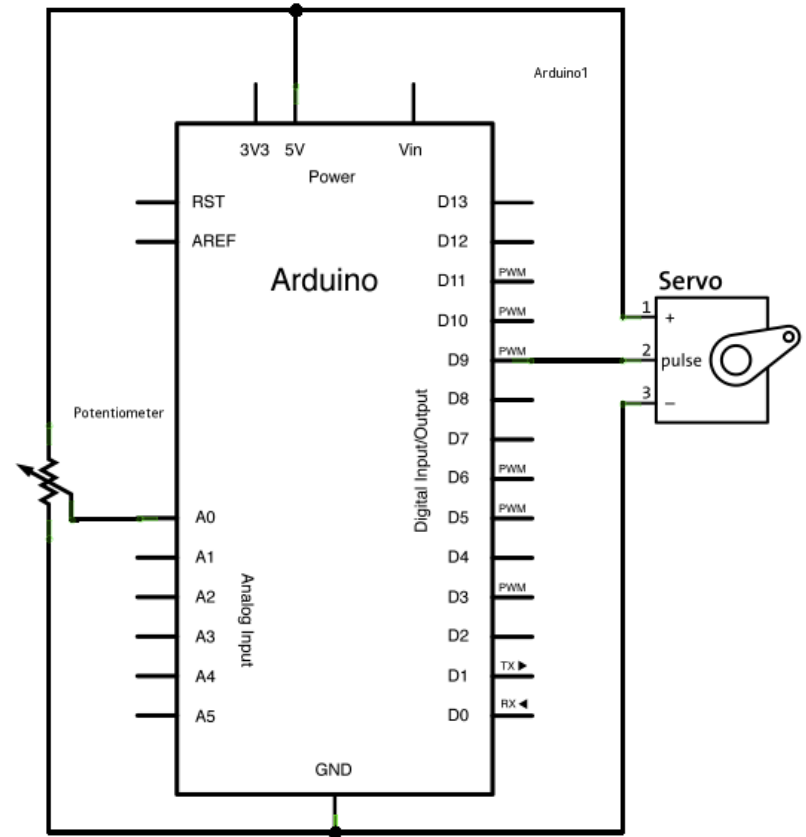
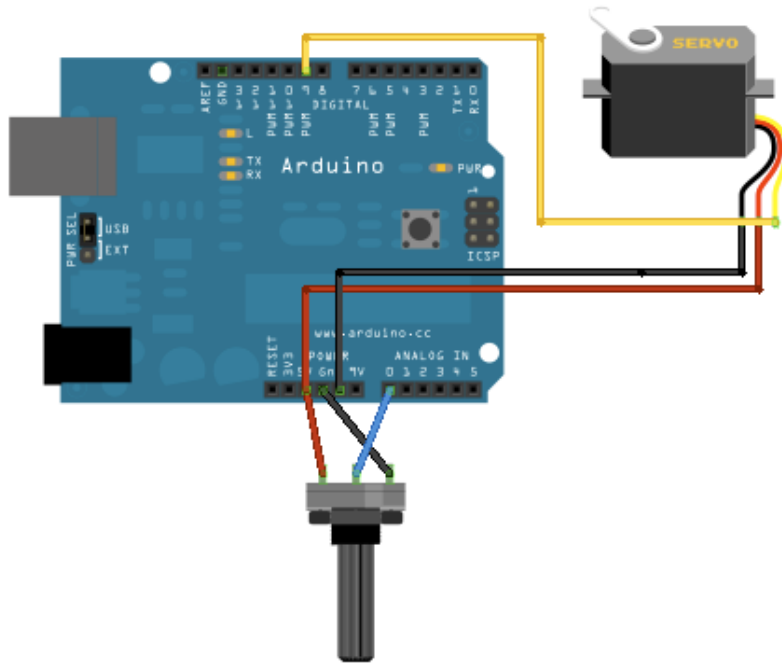
- `pulseServo` sends a pulse of the specified duration and then waits for 15 ms
  - actually, to have 50 Hz, the waiting time should be  $1000/50 = 20\text{ms}$
  - the difference is for taking into account the instruction execution time and the pulse time
    - empirical value
- Empirical tests for this servo
  - pulse duration for -90 is about 250 us (0.25 ms), while +90 is 2250 us (2.25 ms)

# SERVO LIBRARY

- Reference library to control servos: Servo
  - <https://www.arduino.cc/en/Reference/Servo>
  - using PWM
- Library API
  - **attach**(pin)
    - attach the servo to a specified pin
    - on Arduino UNO before 0017 version => 9 and 10 only
  - **write**(angle)
    - to specify the angle of the shaft, from 0 to 180
- Examples
  - “*Knob*”
    - control by means of a potentiometer
    - <https://www.arduino.cc/en/Tutorial/Knob>
  - “*Sweep*”
    - continuous rotation forward and back

# KNOB

- <https://www.arduino.cc/en/Tutorial/Knob>



# KNOB

```
/*
Controlling a servo position using a potentiometer (variable resistor)
by Michal Rinott <http://people.interaction-ivrea.it/m.rinott>

modified on 8 Nov 2013
by Scott Fitzgerald
http://www.arduino.cc/en/Tutorial/Knob
*/

#include <Servo.h>

Servo myservo; // create servo object to control a servo

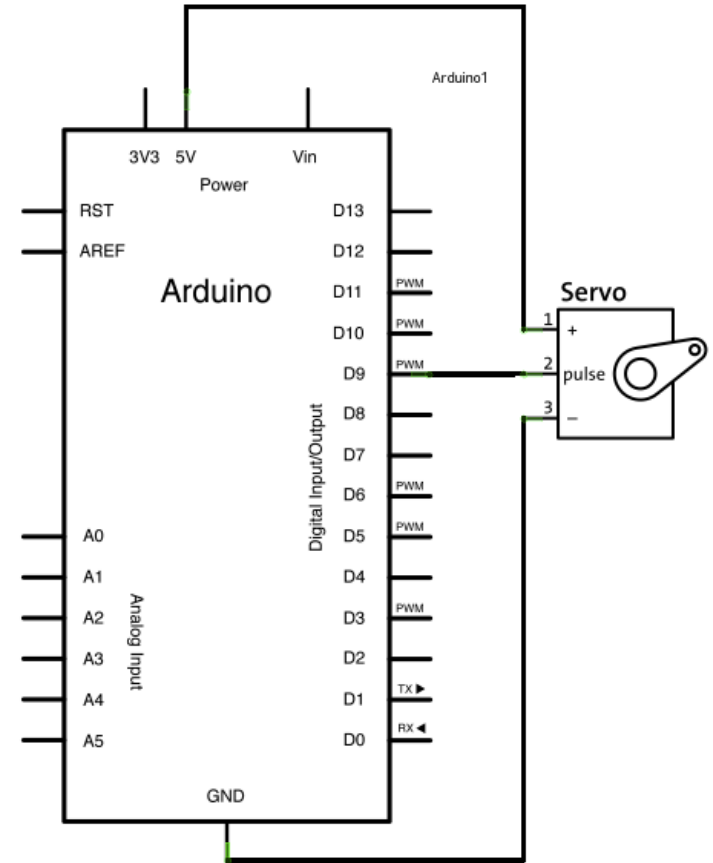
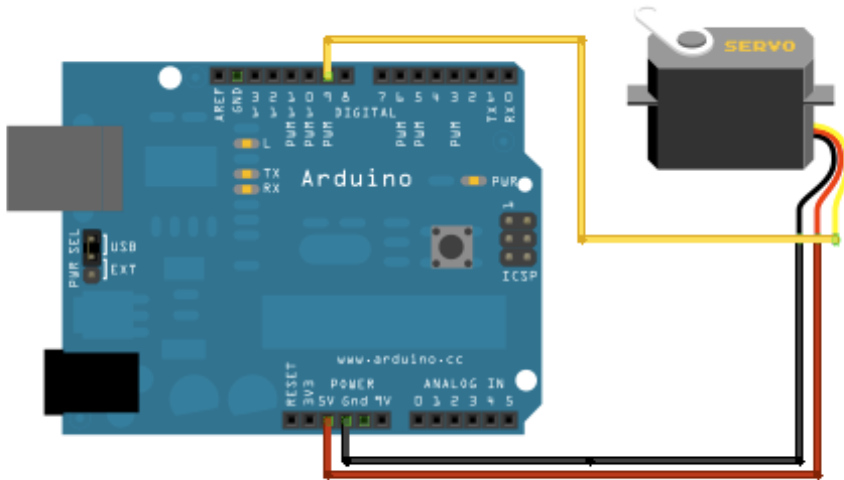
int potpin = 0; // analog pin used to connect the potentiometer
int val; // variable to read the value from the analog pin

void setup() {
 myservo.attach(9); // attaches the servo on pin 9 to the servo object
}

void loop() {
 val = analogRead(potpin); // reads the value of the potentiometer (value between 0 and 1023)
 val = map(val, 0, 1023, 0, 180); // scale it to use it with the servo (value between 0 and 180)
 myservo.write(val); // sets the servo position according to the scaled value
 delay(15); // waits for the servo to get there
}
```

# SWEEP

- <https://www.arduino.cc/en/Tutorial/Sweep>



# SWEEP

```
/* Sweep
by BARRAGAN <http://barraganstudio.com>
This example code is in the public domain.
modified 8 Nov 2013
by Scott Fitzgerald
http://www.arduino.cc/en/Tutorial/Sweep
*/

#include <Servo.h>

Servo myservo; // create servo object to control a servo

int pos = 0; // variable to store the servo position

void setup() {
 myservo.attach(9); // attaches the servo on pin 9 to the servo object
}

void loop() {
 for (pos = 0; pos <= 180; pos += 1) { // goes from 0 degrees to 180 degrees
 // in steps of 1 degree
 myservo.write(pos); // tell servo to go to position in variable 'pos'
 delay(15); // waits 15ms for the servo to reach the position
 }
 for (pos = 180; pos >= 0; pos -= 1) { // goes from 180 degrees to 0 degrees
 myservo.write(pos); // tell servo to go to position in variable 'pos'
 delay(15); // waits 15ms for the servo to reach the position
 }
}
```

# DC MOTORS

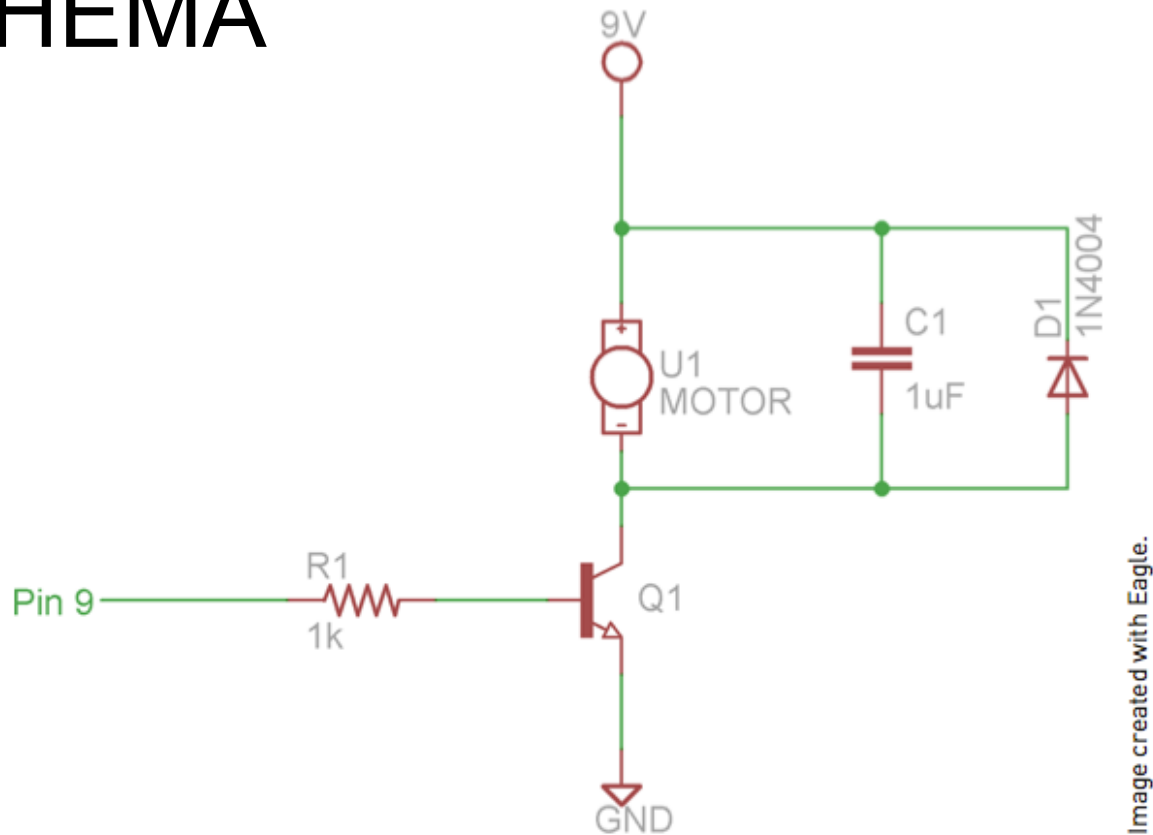
- Two types
  - brushless (stepper)
  - brushed
    - cheaper but last less
- Control => on the speed of the shaft
  - by changing the voltage at the terminals, we change the rotation speed of the shaft
  - by inverting the voltage, we invert the rotation direction
    - use of the H-bridge circuit

# EXAMPLE IN ARDUINO

- In general, a **DC motor needs a separate power line**
  - Arduino does not give enough current to power the motor
  - besides, when operating a motor can generate current spikes in the opposite direction => can damage the micro
  - For these reasons, a separate circuit is used to power the motor DC and Arduino is used only for control
- In the next example
  - the motor is powered by a separate battery
  - the speed control is done by means of a PWM signal
  - a transistor (electronic component) is used as switch

Example taken from[EA, cap. 4]

# SCHEMA

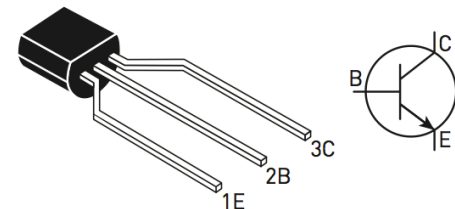


## Components

- 9V battery
- Q1 is a NPN BJT (Bipolar-junction transistor) used as a switch
- 1 KOhm is used to separate the base terminal of the transistor from the Arduino control pin
- U1 is the motor
- C1 is a capacitor used to filter the noise generated by the motor
- D1 is a diode that protects the common power supply of the battery from invert current generated by the motor

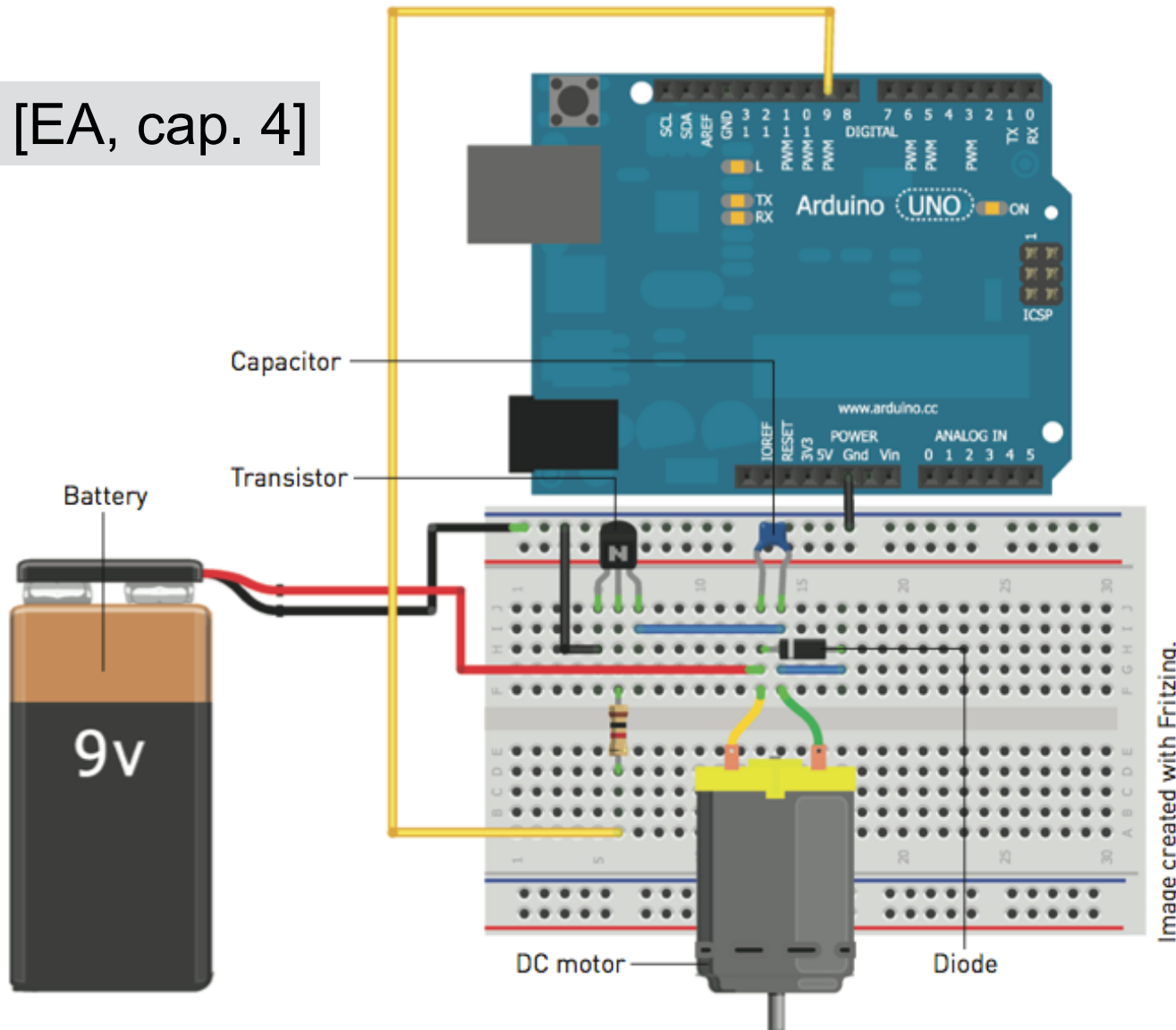
# HOW IT WORKS

- A transistor is used to modulate the speed of the motor by means of a PWM signal switches on/off the power
  - Transistors are widely used in electronics for many different purposes
    - amplifiers, logical ports, switches
  - The current flows from the collector to the emitter of the transistor
  - Modulation occurs on the base B pin
    - when a specific voltage is applied to B, the circuit is closed and the current flows, the shaft moves
    - if/when the power is stopped abruptly, then motor energy is dissipated as a voltage peak generating current in the opposite direction
      - for this reason a protection diode (diode of recycle) is used
- The PWM duty cycle => speed of the motor
  - duty cycle = 100% max speed
  - duty cycle = 0% speed = 0
  - linear pattern



# SKETCH

[EA, cap. 4]



## CHECKS | WARNING

- same common GND must be used
- 9V should **not** be connected to 5 V
- both the transistor and the diode must be oriented in the proper way

Image created with Fritzing.

# PROGRAMMING

- Speed control through a PWM signal with a specific duty cycle

```
const int MOTOR = 9;

void setup()
{
 pinMode(MOTOR, OUTPUT);
}

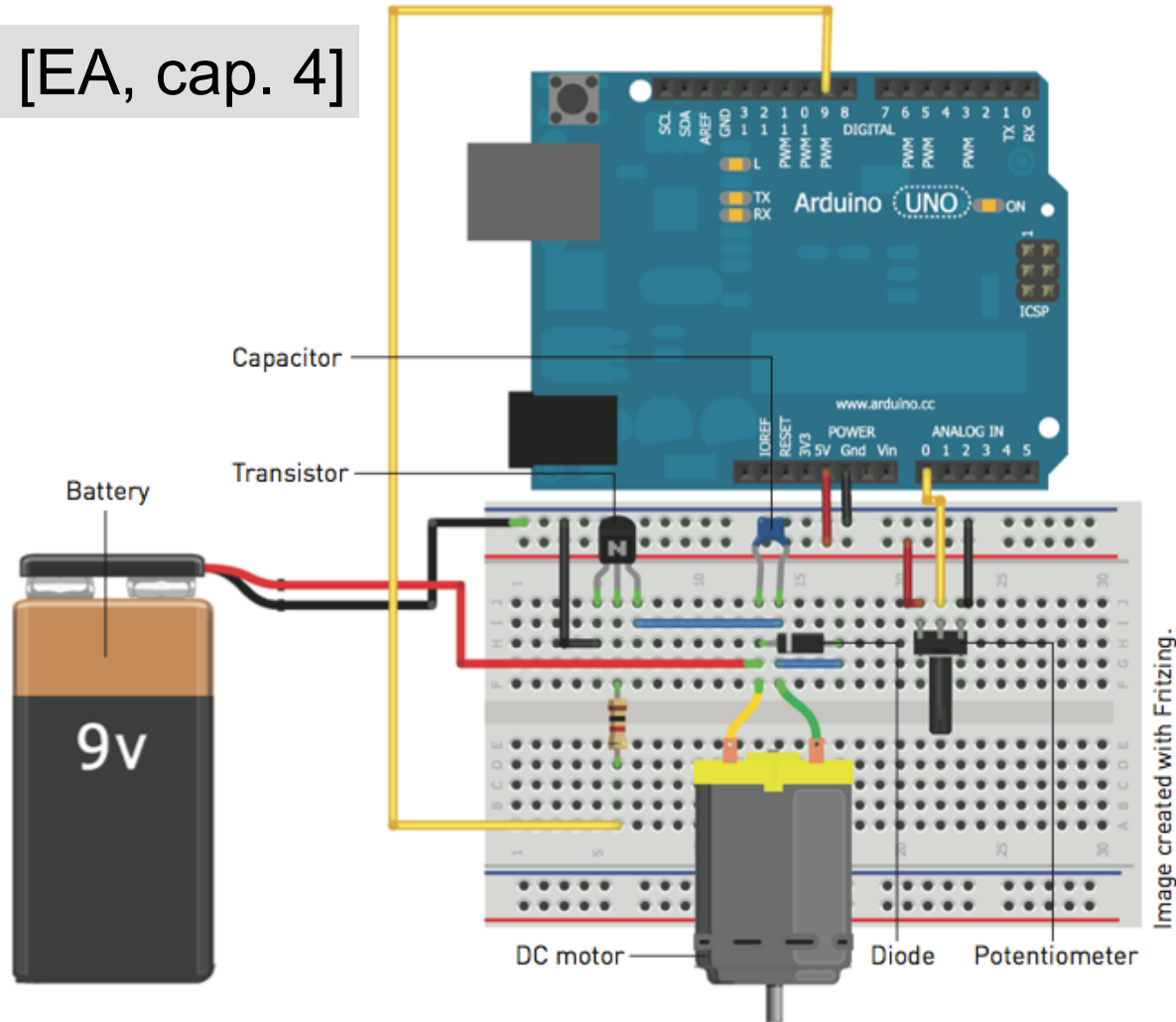
void loop()
{
 for (int i = 0; i < 256; i++)
 {
 analogWrite(MOTOR,i);
 delay(10);
 }

 delay(2000);

 for (int i = 255; i >= 0; i--)
 {
 analogWrite(MOTOR,i);
 delay(10);
 }
 delay(2000);
}
```

# ADDING A POTENTIOMETER

[EA, cap. 4]



```
const int MOTOR = 9;
const int POT = 0;

int val = 0;

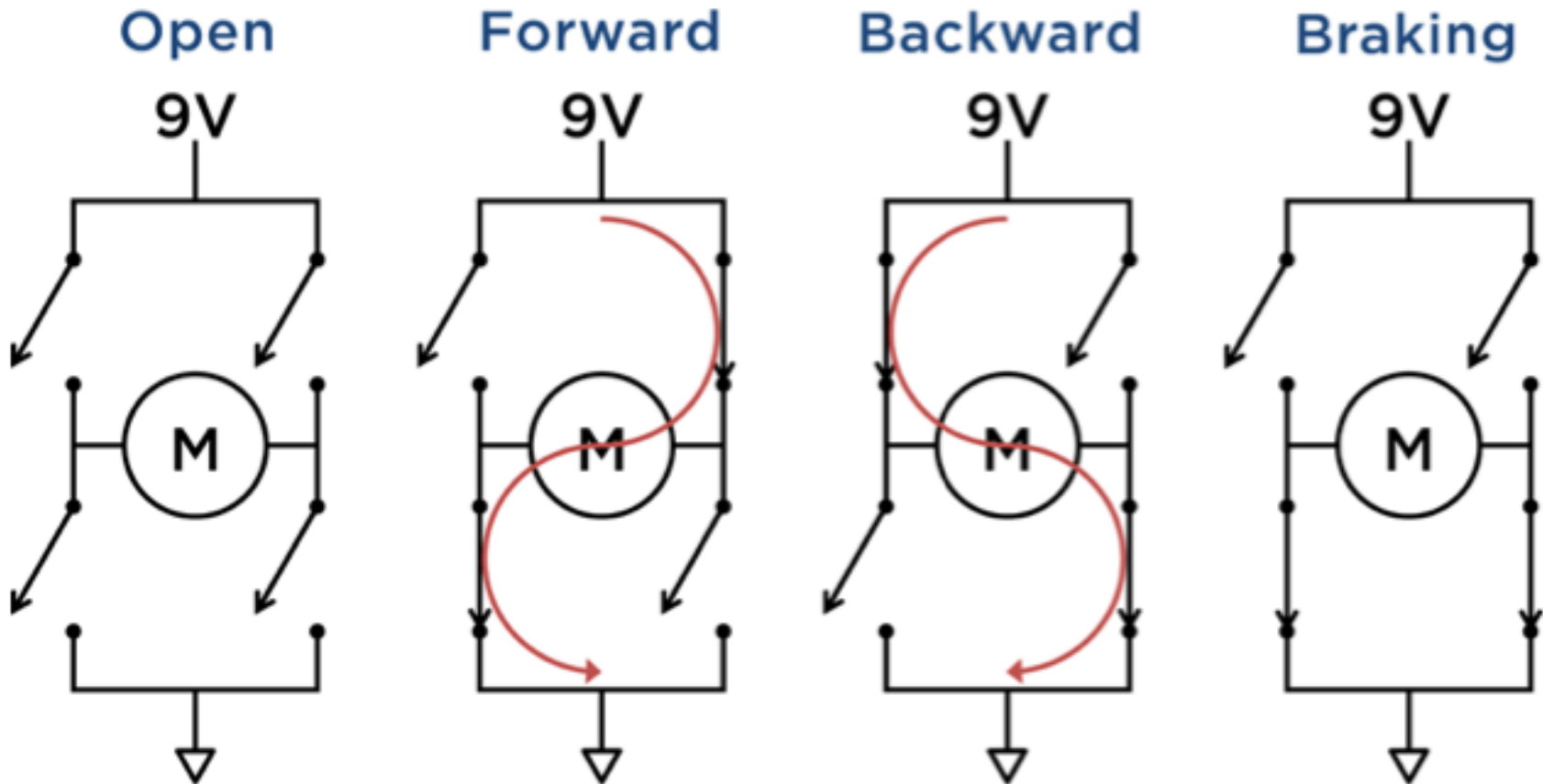
void setup()
{
 pinMode(MOTOR, OUTPUT);
}

void loop()
{
 val = analogRead(POT);
 val = map(val,0,1023,0,255);
 analogWrite(MOTOR,val);
}
```

# H-BRIDGE FOR DIRECTION CONTROL

- It is possible to change movement direction by inverting the voltage at the terminals
- An H-bridge component is used for this purpose.
  - it integrates 4 switches (realised by transistors) and all the protection circuits
- 4 operational states
  - **open**
    - all switches open and motor is off
  - **forward**
    - 2 switches are open on the opposite diagonal, the current flows and the motor rotates in one direction
  - **backward**
    - the other 2 switches are open, the current flows in the opposite direction, the motor rotates in the opposite direction
  - **braking**
    - slowing down the motor until stop

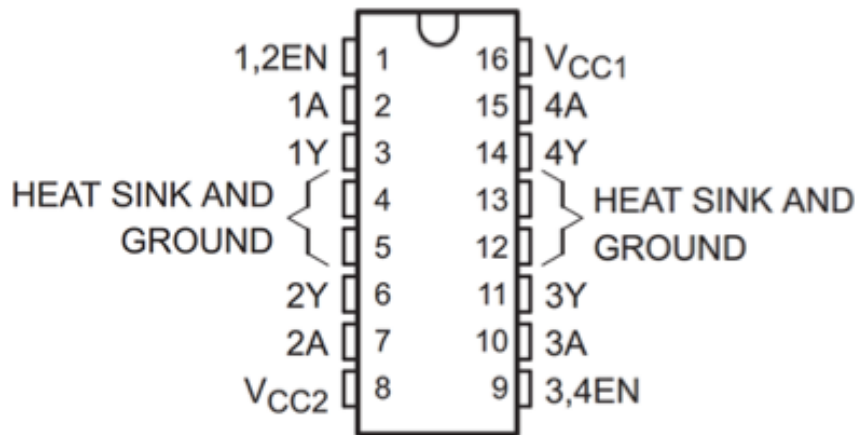
# H-BRIDGE OPERATING STATES



[EA, cap. 4]

# H-BRIDGE: EXAMPLE

- Example: SN754410 Quadruple Half-H driver
  - can drive 2 motors



FUNCTION TABLE  
(each driver)

| INPUTS <sup>†</sup> |    | OUTPUT |
|---------------------|----|--------|
| A                   | EN | Y      |
| H                   | H  | H      |
| L                   | H  | L      |
| X                   | L  | Z      |

H = high-level, L = low-level

X = irrelevant

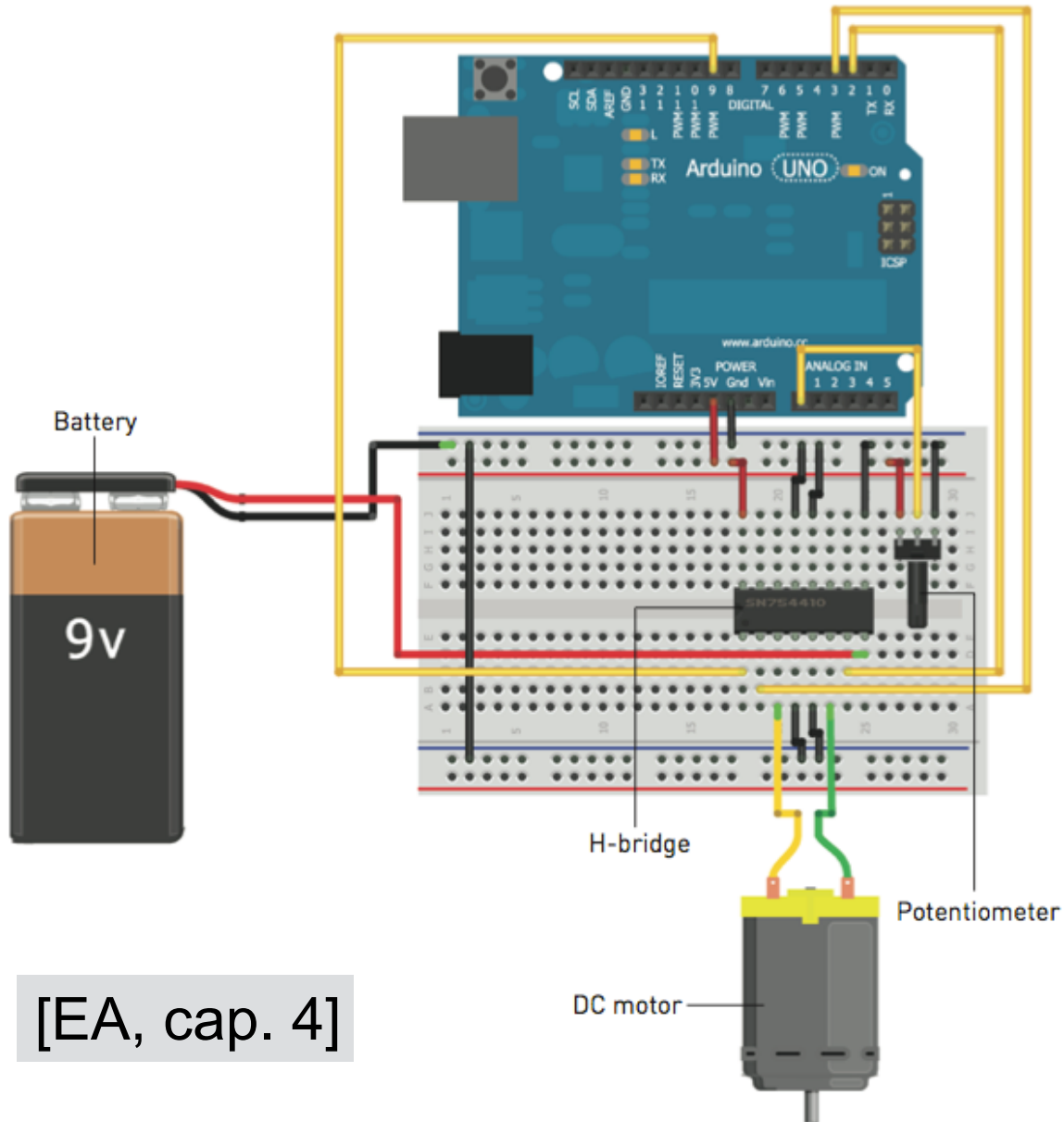
Z = high-impedance (off)

<sup>†</sup>In the thermal shutdown mode, the output is in a high-impedance state regardless of the input levels.

Image used with permission courtesy of Texas Instruments.

- VCC2 => motor power line - 9V
- VCC1 => Arduino/logic circuit power - 5V
- 1Y e 2Y - output for left motor
- 1A e 2A - switch state for left motor, to be wired to Arduino I/O pins
- 1,2EN - pin to enable/disable left motor - to be wired to PWM Arduino pin
- 3Y e 4Y - output for right motor
- 3A e 4A - switch state for right motor, to be wired to Arduino I/O pins
- 3,4EN -pin to enable/disable right motor - to be wired to PWM Arduino pin

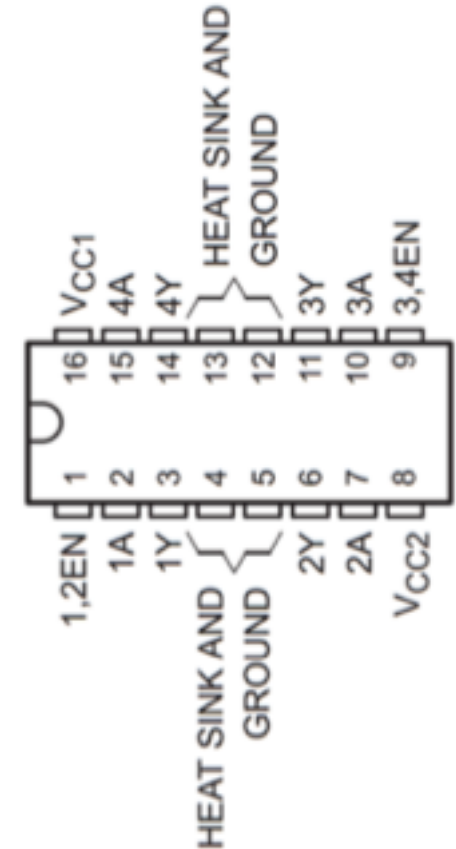
# SKETCH WITH H-BRIDGE



[EA, cap. 4]

Image created with Editings

JATORS



# PROGRAMMING

- Using a potentiometer, to change the rotation from one direction to the other one

```
const int EN = 9; // Half bridge1 Enable
const int MC1 = 3; // Motor Control 1
const int MC2 = 2; // Motor Control 2
const int POT = 9; // POT on Analog pin 0

int val = 0; // memorizza valore potenziometro
int velocity = 0; // velocità desiderata (0-255)

void forward(int rate)
{
 digitalWrite(EN, LOW)
 digitalWrite(MC1, HIGH)
 digitalWrite(MC2, LOW)
 analogWrite(EN, rate)
}

void backward(int rate)
{
 digitalWrite(EN, LOW)
 digitalWrite(MC1, LOW)
 digitalWrite(MC2, HIGH)
 analogWrite(EN, rate)
}
```

```
void brake()
{
 digitalWrite(EN, LOW)
 digitalWrite(MC1, LOW)
 digitalWrite(MC2, LOW)
 digitalWrite(EN, HIGH)
}

void loop()
{
 val = analogRead(POT);

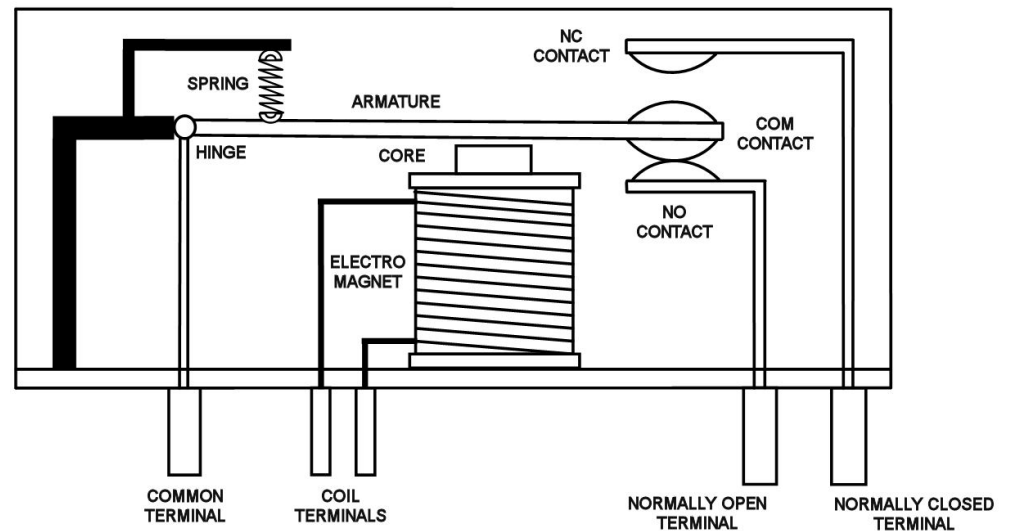
 if (val > 562){
 velocity = map(val, 563, 1023, 0, 255);
 forward(velocity);
 } else if (val < 462) {
 velocity = map(val, 0, 461, 0, 255);
 forward(velocity);
 } else {
 brake();
 }
}
```

# CONTROLLING EXTERNAL CIRCUITS

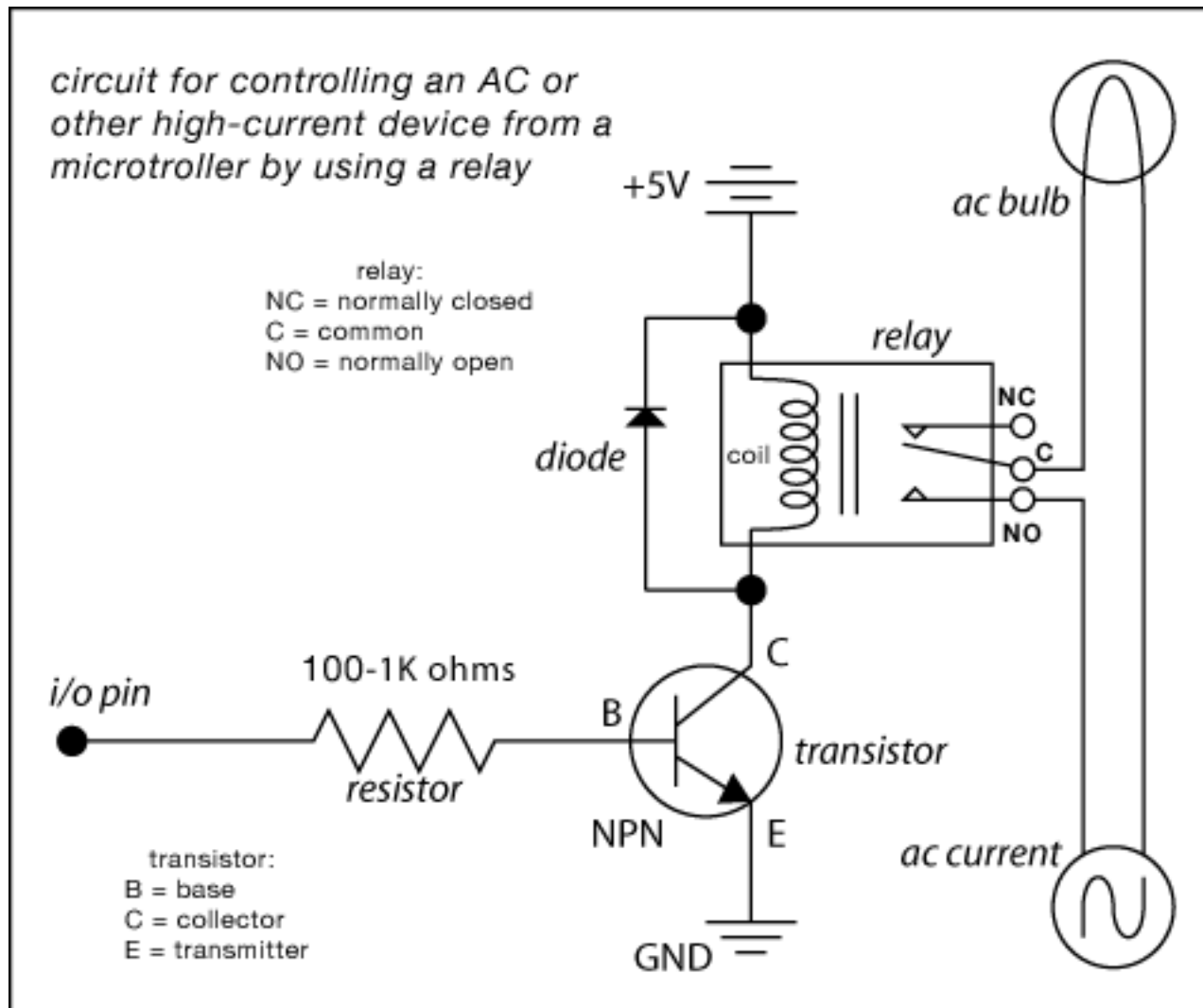
- Microcontrollers can be used to external circuits, powered with a different power — including alternating current circuits (AC)
  - To this purpose we need electronic devices functioning as switches, like transistors
- Electronic components functioning as switches:
  - **relay**
    - open/close a circuit by means of an electromagnet, that can be controlled by the microcontroller
    - Example in Arduino
      - <http://www.instructables.com/id/Arduino-Controlled-Relay-Box/>
  - **optocoupler**
    - based on an internal led, controlled by the micro
    - when the led is switched on, a photocell is activated and the second circuit is closed

# RELAY SWITCH

- Internally it has 2 circuits — the one with small current (where the micro is connected) and the one with big voltage and currents (e.g. 220 V AC)



# RELAY WIRING



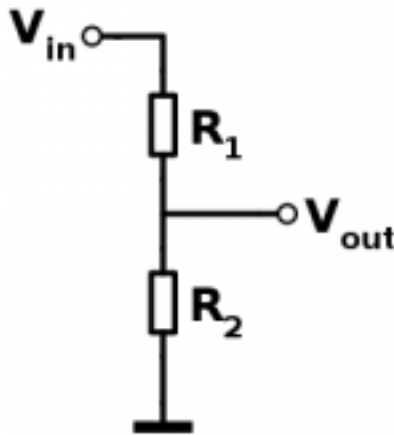
# ABOUT INTERFACING COMPONENTS USING DIFFERENT VOLTAGE LEVELS

# DIFFERENT VOLTAGE LEVELS PROBLEM

- Sensors/actuators can operate at 2 different voltage levels
  - 5 V (e.g.: Arduino) and 3.3 Volt (e.g. ESP32, Raspberry PI,..)
- In order to interface/use sensors/actuators that work at 3.3V into MCU boards that work at 5V we need to adopt specific strategies not to damage the components
- This is true also when connecting systems through serial buses
  - e.g. by connecting an Arduino and ESP by means of the asynchronous serial or the synchronous one
  - for the asynchronous serial
    - TX (ESP) => RX (Arduino): no problem
      - 3.3V level is enough to be interpreted as HIGH in Arduino
    - **TX (Arduino) => RX (ESP): problem!**
      - 5V in input for ESP can damage the board
- Solutions
  - using **voltage dividers**
  - using **bi-directional logic level converter**

# VOLTAGE DIVIDER

- It can be used to produce an output voltage  $V_{out}$  (e.g.: 3.3V) starting from a higher voltage  $V_{in}$  (e.g. 5V)
- The clue is the ratio between the resistances (not their absolute values)
- However, the absolute value impacts on the circuit current
  - $I_1 = I_{out} + I_2, I_{out} = I_1 - I_2$



$$\alpha = \frac{R_2}{R_1 + R_2}$$

$$V_{out} = \alpha V_{in}$$

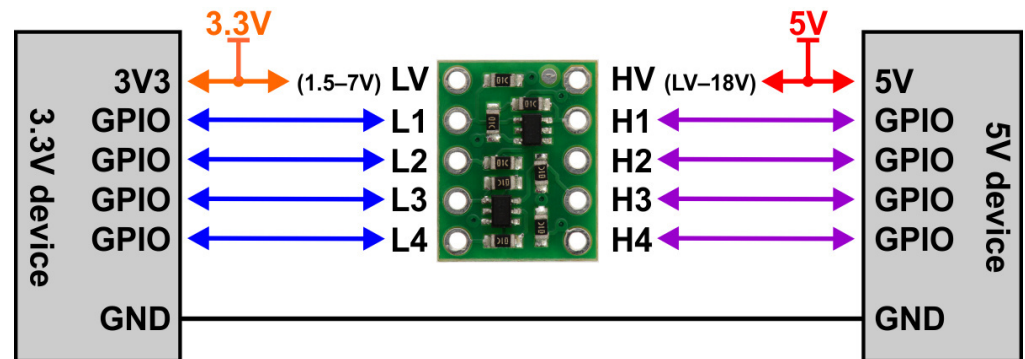
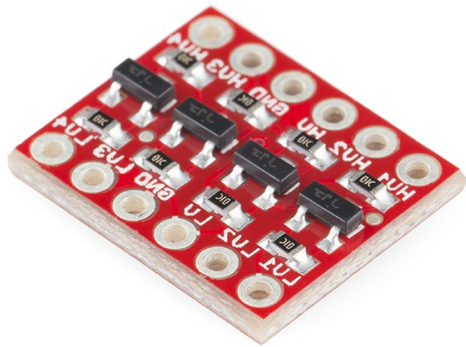
Typical value used:

**R1** = 1 KOhm

**R2** = 2 KOhm

# BI-DIRECTIONAL LOGIC LEVEL CONVERTER

- The Bi-Directional Logic Level Converter (LCC) is a module that reduces the voltage from a signal.
  - recommended method to level shift signals that require high-speed communication (e.g. synchronous serials such as I2C...)



[<https://randomnerdtutorials.com/how-to-level-shift-5v-to-3-3v/>]

<https://community.blynk.cc/t/using-logic-level-converter-to-connect-esp-and-i298/15323>

# REFERENCES

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- Fondamenti della Misurazione. E.Bava et al. Esculapio
- [EA] *“Exploring Arduino”*, Jeremy Bloom, Wiley
- *“Make: Sensor - Projects and Experiments to measure the world with Arduino and Raspberry Pi”* (Karvinen et al). Maker-Media